How far does a learned surrogate of a flowsheet simulator transfer across ores? A closed-circuit, size-by-mineral engine, with cases and ore groups held out

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Abstract

Learned surrogates of process simulators are often evaluated on random states from simulated plants they have already seen. We compare interpolation, leave-one-case-out and leave-one-ore-group-out evaluation for five models trained on a steady-state mineral-processing simulator. The simulator tracks mineral mass over 63 size classes, water and closed grinding circuits, with unit-level balance audits and explicit misses against published examples. On a 3,072-state design over twelve authored cases, the multilayer perceptron has an interpolation recovery RMSE of 3.38 percentage points. Its magnetite transfer forecast is wholly outside a declared gross-range diagnostic and remains a rejected fold with null error metrics. Over the eleven case folds admissible for all models, mean recovery RMSE is 6.86 points for the perceptron, 8.67 for gradient boosting and 9.40 for the Gaussian process. These conditional results do not describe performance on the excluded case. Four of five neural seeds reject magnetite; the remaining seed retains a 217.18-point error, exposing instability that an interpolation score conceals. The perceptron's recovery R^2 is 0.795 on five chalcopyrite cases held out together and -10.2 on two gold-in-pyrite cases. Group-held-out log-upgrade median R^2 is negative for every model. Results therefore depend on the target, held-out unit, admissibility diagnostic and cohort denominator. The cases are authored; these results do not establish performance on operating plants.

1 Introduction

A surrogate of a flowsheet simulator is attractive for the same reason as any surrogate: the simulator is slow and the surrogate is fast, so it can sit inside an optimizer, a sweep or an interactive tool. Whether it can stand in for the simulator depends on the question it is asked. Inside the envelope it was trained on, an interpolation score answers that question. A digital twin is asked something harder: what happens with a different ore, a harder or finer feed, another plant. That is transfer, and an interpolation score says nothing about it.

This work measures both on the same models, with a simulator whose answers are known exactly and can be reproduced. Its contributions are three:

1. an open flowsheet engine that represents streams by size and mineral, solves closed circuits, audits every balance from its output streams, is checked against published examples with every miss stated, and is reproduced by a line-by-line second implementation;

2. a test protocol that scores learned surrogates of it by interpolation, by leaving one case out and by leaving one ore group out over twelve authored plants in four circuit families, with interval coverage and an out-of-envelope guard;
3. a measured transfer failure and seed instability on magnetite, conditional comparisons that retain the failed fold, distinct results for whole-case and ore-group holdouts, and a reconstruction guard whose missed feature shifts remain visible by input and distance.

Published comminution and simulator studies provide context for the controlled engine comparison; their plant-specific results are not transferred as OreFlow calibration data [31, 30].

Grinding and flotation couple particle-size transformation to classification, water circulation and mineral-specific recovery. Finer grinding can improve liberation while increasing energy and changing the flotation feed. A control selected from one unit response may therefore violate a downstream constraint.

OreFlow exposes these couplings through stream and unit records. It creates a controlled teacher for method comparison and a transparent setting for testing how validation units alter conclusions; it does not replace plant sampling or metallurgical accounting.

The report distinguishes numerical closure within implemented equations, reproduction of external reference cases and predictive agreement on held-out records. Passing one category does not imply passing another.

2 Problem formulation

Let c index one of twelve authored circuit cases and let v index one of its declared operating variants. OreFlow evaluates the map

$$z = \mathcal{E}(c, u; \theta),$$

where u is the case's operating point and θ contains its mineralogy, size distribution, liberation, density, hardness and equipment declarations. The output z contains size-by-mineral streams, water flows, equipment loads and derived assays. A call returns a converged steady state or a structured refusal. Refusal is a first-class outcome: it is not a zero recovery and cannot be silently omitted from a denominator.

For mineral m, unit feed rate F_m and output rates $O_{m,k}$, conservation requires

$$r_m = F_m - \sum_k O_{m,k} = 0.$$

Water has a separate balance. Recovery is the payable mineral in the selected product divided by payable feed; grade is elemental payable mass divided by total concentrate solids. Because the two denominators differ, they are not interchangeable. Increasing mass pull can raise recovery while lowering grade. The engine checks the balances at unit boundaries, then aggregates the case record only when its invariants pass.

The forward question is to compute the steady state for an admissible case and operating point. The learning question is to approximate a vector of outputs from a feature vector, with recovery, log upgrade and specific energy as separate targets. Its estimand is not simply "accuracy": interpolation within a known case, transfer to an entire held-out case, and transfer to a held-out ore group define distinct deployment questions. The optimization question chooses bounded controls to maximize normalized recovered metal and energy subject to grade, installed-power and water limits. The uncertainty question propagates declared ranges in work index, head grade, liberation size and floatability through the same engine. None of these quantities is an economic value: OreFlow has no prices, reagent costs, capital model or production schedule.

This formulation also marks what the product does not claim. $\mathcal{E}$ is steady state, not a start-up or control-loop model. Its authored parameters are not calibrated to operating-plant observations. An optimum is a finite-search feasible candidate from the declared equations; it is not an operating set point. The externally measured datasets have their own lanes and protocols. They neither alter the authored cases nor validate

their circuit parameters. These separations are necessary to interpret every downstream score and to avoid promoting a modeled output into a plant measurement.

3 Notation

Bold capitals denote size-class operators and vectors. Scalar rates are in tonnes per hour unless otherwise stated; the case record retains the exact field unit. Indices identify mineral m, size class i, equipment unit j and sampled state n. The same letter can denote different domain objects only where its unit and subscript make that role explicit: F_m is a mineral mass rate, while F in the Bond equation is a characteristic feed size. The probability operator is written Pr; installed power is P_{inst}.

Symbol	Meaning	Unit or domain
c, v	authored case and declared variant	identifiers
u, θ	operating controls and case parameters	mixed; each field has its schema unit
$z, \mathcal{E}$	steady-state streams and exact engine map	state and deterministic map
$F_m, O_{m,k}$	mineral feed and unit-output rates	t/h
r_m	mineral balance residual	t/h
$\mathbf{B}, \mathbf{C}, \mathbf{T}$	breakage, classification and mill operators	dimensionless matrices
d_{50c}, P_{80}	corrected classifier cut and product size	µm
W_i, E	work index and specific energy	kWh/t
R, M, G	recovery, recovered payable and concentrate grade	%, t/h, %
$x, y, \hat{y}$	feature, target and model prediction	declared target units
T_q, n_q	held-out rows and count under protocol q	set and count
Δ, h	normalized search mesh and constraint violation	dimensionless
S_i, S_{T_i}	first-order and total sensitivity indices	dimensionless

The notation separates a requested setting from a solved state. The input vector u is what the user or a study requests; the result z is what the engine converges to. In a power-limited grind, requested P_{80} and achieved P_{80} differ, and only the latter belongs in a size-based energy equation. Likewise, the optimizer's normalized decision coordinate is not a physical control until it is mapped back through the case contract. These distinctions matter in figures: an axis labelled with a target must not be read as an observed outcome.

The index q denotes an evaluation protocol, not a model. Interpolation, case holdout and grouped transfer define different training and test sets, so T_q is retained in equations to make the denominator explicit. The sample count n_q changes after a rejection, while the attempted count remains fixed. A score computed from only admitted rows is conditional on that admission rule. In the process methods, i can index size classes and j can index flotation cells; the corresponding vector dimensions make the intended summation clear. Bold operators act on the size grid and are applied separately for every mineral before stream totals are formed.

E is used for specific energy in kWh/t, while e denotes mill energy per pass on the internal mill-feed basis. Their denominators differ. M is recovered payable rate in t/h, not total concentrate mass. G is grade on the stated elemental basis, not mineral mass fraction unless the case defines it so. The report names recovery units as percentages, and any difference in recovery is a percentage-point difference. These conventions prevent common scale errors when comparing kinetic percentage recovery, a mass-flow response and a dimensionless model score.

Dimensionless normalized objectives are used only inside the optimizer; displayed process metrics retain physical units. Recovery is reported in percent, while the difference between two recoveries is in percentage points. Specific energy is power divided by solids throughput and is not interchangeable with total power. Water intensity is volume per solids mass, whereas water flow is a volume rate. These distinctions keep throughput changes from being mistaken for changes in process water demand. Subscripts identify the stream, mineral, size class or sampled state when the role is not already explicit. The conventions are carried into schema units, serialized traces and table captions.

4 Data and data contracts

The twelve case definitions are project-authored scenarios under the repository's MIT licence, not a measured plant dataset. Each definition identifies process family, category, scientific question, provenance, sources, KPI ranges, nominal controls and eight variants. The committed 0.10.010 bake contains 96 variant records generated with contract digest `0d2b65d272f3abcc6adffbf0f734c985cb002ab5cb665c109b8e8e9e31cecbd7`. The learning design contains 3,072 simulator states, 256 per authored case. Labels are exact relative to the implemented engine. They are not ground truth about an operating mill.

The catalog in `docs/contracts/catalog.md` is the human-readable authority for every contract's identifier, version, schema, producer, consumer, missing-data behavior and status. The operating contract supplies units, bounds, applicability and refusal rules. Case and manifest contracts bind authored inputs and solved outputs. Benchmark and studies contracts bind comparisons; learning, surrogate and screen contracts bind feature order, target order, model identity, scaling and eligibility decisions. Probe and validation contracts preserve boundary challenges and checks. Source contracts keep the GeoMet LCT, HZDR particle, iron-plant soft-sensor and other records distinct by provenance and use. A missing required artifact blocks its dependent result; it is never replaced with a default value. A failed steady state is retained with its refusal code and denominator.

The learning data use three separate held-out protocols. Interpolation divides rows within case; case transfer withholds all rows of a case; group transfer withholds related ore cases together. Every scaler is fit only on the training portion of each fold. The measured-data lanes use source-specific records and splits. The iron-plant lane, for example, groups consecutive hours and applies a temporal embargo; it does not feed measurements into the circuit engine. The locked-cycle GeoMet data lack grind, reagent and residence controls, so they cannot identify all parameters that the model would need for calibration. Field names that resemble operating controls do not make the missing measurements available.

For auditability, schema validation and physical validation are separate. JSON schemas check shape, types, required values and enumerations. Domain gates check conservation, allowed ranges, engine identity and cross-field invariants. Producer and consumer tests exercise the same record from creation to use, while adversarial tests change nested boundaries and malformed identities. A contract digest binds the operating definition used in the bake, and each derived artifact carries its engine version. This lets the reader distinguish a schema-valid record from a physically accepted state and from an independent empirical observation.

4.1 Authored case, manifest and aggregate evidence contracts

The case definition is the upstream scientific input: it binds the case identifier, process family, descriptive question, provenance, cited KPI ranges, nominal controls and labeled perturbations. The manifest is its run-level companion. It identifies the engine version and operating-contract digest, the exact artifact file, its variants, nominal result and KPI summaries. A manifest therefore cannot be interpreted as a new independent

observation: it is an index to a named deterministic bake. The aggregate index collects case and variant counts and links the case-level records; consumers use it to discover the committed suite without duplicating every stream array.

The benchmark record has a different role. It groups selected oracle comparisons, kinetic fits, optimizer results, uncertainty outputs, learning metrics and lane summaries under one engine identity. Its fields summarize the scientific comparison; detailed per-case data remain in their child records. This separation lets a reader inspect one reported number and then follow the record that produced it. The studies record collects the switchable studies by case, including method-specific diagnostics and their denominators. The five artifacts are produced by distinct pipeline stages and consumed by validation, the web workbench, notebooks and release checks. If a case record or engine digest is missing, the aggregate stage stops or omits only the dependent section with an explicit status; it does not interpolate a missing case. Version strings and digests must agree before cross-case aggregation, because mixing engine revisions can turn a code change into apparent ore variability.

These contracts carry authored provenance, not measurement provenance. `case_id`, category and family explain grouping; the localized title and question identify the teaching problem; `sources`, `kpi_sources` and `provenance` distinguish literature ranges from project assumptions. `nominal` and `variants` declare what changed between solves. The output record stores the result and the methods tied to that setting. To extend the suite, add a new case with explicit source or "authored" status, validate its operating point against the canonical bounds, run all declared variants, and produce the manifest from the same engine version. A benchmark summary should be regenerated only after each child artifact passes schema and physical checks. This prevents stale indexes from implying that every case was recomputed.

4.2 Operating-input and validation contracts

The operating contract declares the admissible fields, units, bounds, case applicability and refusal behavior used by the engine and optimizer. It is the boundary between user controls and numerical methods. A control outside its bound is not silently clipped: the engine returns a structured refusal that keeps the requested value and reason visible. The nested validation checks ensure that a valid top-level point cannot hide an invalid nested decision. Browser and Python consumers share this declaration, and parity tests check that both accept or refuse the same boundary values.

The contract-probe record is deliberately adversarial. It stores a digest of the contract plus probes that test lower and upper edges, just-outside values, wrong types, missing fields and cross-field combinations. These probes are test evidence, not operating scenarios. Their expected response is compared with the implementation so a UI can distinguish a disabled control, a refused replay and a numerical result. The validation record summarizes whether stages passed, the stage-specific diagnostics, error list, worker count and elapsed-time groups. A `passed` flag is an aggregate status; it does not replace the individual errors or prove scientific validity of model assumptions.

The validation artifact is produced after the bake and consumed by release readiness, CI records and documentation. A missing probe or validation file blocks the corresponding gate. A test must fail when a required boundary changes without updating the contract; accepting a changed schema merely because the nominal case still runs would leave the contract drift undetected. `contract_digest` is computed from canonical contract content and tied into downstream artifacts. Thus a case record with the right schema but an obsolete contract digest is not an equivalent input to a newly declared optimizer. The reviewer should inspect both the declared domain and the refusal evidence before interpreting model coverage.

4.3 Learning design and model-inference contracts

The learning design record defines feature order, target order, model names, sample design, split identity, settings and the interpolation, case-held-out and group-held-out results. Its fold arrays preserve attempted, scored and rejected populations. The earlier schema version remains declared for historical records, while the current version carries the expanded identities and transfer diagnostics. A consumer must not treat those versions as interchangeable: it checks the `schema` value and the expected fields before interpreting a score.

The case group is not supplied as a predictive feature in the transfer protocol.

The process-surrogate record binds the numerical inference transform. It stores ordered features and targets, feature means and scales, target means and scales, separate reconstruction-guard scaling, guard threshold and reference ranges. These arrays are positional: changing feature order without changing the record could produce finite but meaningless predictions. The screen record binds the exported network and GP metadata, covariance factor identity and the references used by the optimizer's preflight. Inference checks model and contract identity before evaluating a candidate. The solver uses the neural response only to rank proposals; exact-engine evaluation remains the source of objective and constraint values.

The learning stage produces the design and fold results; training and export produce the model records; the notebook and workbench consume them. Artifact hashes, engine version, contract digest, target order and declared thresholds let the consumer reject a stale or incomplete export. Missing scaler values, a wrong target count, non-finite model weights or a mismatched GP factor are hard failures. They are not filled with identity scaling or a broad interval. The frozen-model challenge records attempted and accepted samples so guard sensitivity can be inspected without deleting misses. Its known-case seed is a reproducibility design, not a new-ore test. A prospective plant lane would require a separate contract for time, sampling, assays and production-period splits.

4.4 External datasets and measured-source contracts

External-data interpretation retains the original sample and source identity, including the gravity-recovery sources and the stated particle benchmark protocol [15, 16, 21]. The Geomet locked-cycle record holds source identity, sampling protocol and the paired assay/recovery rows used for whole-hole and spatial-zone evaluation. It retains which measurements are missing: grind, collector, residence and plant circuit controls are not inferred from the assay columns. The iron-plant soft-sensor record stores source and data-quality descriptions, temporal split protocol, pooled scores, model comparisons, repeated-label share, held-label definition and fold results. The embargoed future windows and repeated laboratory values must be read together; the raw row count is not the effective independent sample count.

The HZDR summary declares the dataset DOI, source URL, licence, feature groups, row counts and whether the raw file is locally available. The particle benchmark stores the raw source identity, protocol, standardization, fitted sparse-logistic coefficients and results per constructed case. Test-set probabilities and author predictions are not observed particle labels, so agreement with them is a comparison to a constructed reference. The real-samples record preserves the source, label meaning, assumed floatability ratios, plant or laboratory context, comminution linkage, excluded rows, per-sample estimates and sensitivity summaries. The GeoMet assays are not direct controls for the authored circuit; the record therefore keeps the transformation and assumptions explicit.

The GeoMet, HZDR, iron-plant and real-sample artifacts are produced by separate preprocessing or evaluation stages. Their consumers include research notebooks and the report, not the authored case generator. Each source contract preserves its licence, stable source reference, row exclusions and protocol boundary. A missing raw source or changed file digest blocks reproduction of that lane while leaving independent simulator records valid. It must not be replaced with a synthetic row under the external-source label. Conversely, a source record that is available and schema-valid still needs its study-specific leakage, provenance and measurement checks before a result can be described as measured evidence. This explicit lineage is what allows the report to compare different evidence types without merging their denominators.

5 System architecture

The pipeline is an artifact chain. Case definitions and the operating contract enter preprocessing; the steady-state engine solves the closed circuit and emits streams, equipment values and balance diagnostics. The cases stage writes each nominal and variant record. The benchmark stage compares selected results with published worked examples and retains discrepancies. Separate stages build the particle, source-sample and studies records. The learning stage creates a seeded design, fits the declared models under the split protocols,

and exports scalers, target metadata, neural weights and the GP factor used by inference. Every stage uses versioned inputs and emits its contract identity; no stage fills a missing prerequisite by inventing a score.

Python is the research reference implementation. The TypeScript engine is a second implementation used for the interactive workbench and is checked against the same committed cases. That parity check detects translation errors but is not independent validation of the physical model, since the equations and authored constants are shared. The client reads baked data for reported evidence and runs local calculations for explicit user changes. The optimizer evaluates accepted candidates through the exact process engine; the learned models may prioritize proposals but do not supply final objective values. The app has no plant telemetry connection.

The UI therefore exposes three value origins: a committed result, a local engine replay, and a learned estimate. The provenance, active case, controls and replay status accompany the relevant view. Focus mode changes available screen area and navigation while preserving those identities. A browser-generated state is still a result of authored steady-state assumptions; interactive controls do not make it a plant recommendation. This architecture is intended to make the method inspectable: controls feed a named call, contracts constrain its domain, artifacts bind reproducible comparisons, and the interface presents the resulting state without implying an unrecorded source.

The artifact flow is summarized in Figure 1: versioned inputs pass through the exact engine and stage-specific validations before reaching the evidence records and the browser workbench. Arrows indicate handoffs between versioned artifacts, not a live plant data feed.

A pipeline stage owns the artifact it writes, and consumers validate the schema instead of reaching into an upstream stage's private implementation. This arrangement makes a method replaceable without changing the meaning of its contract. For example, a new kinetic fit may add diagnostics to its study record, but it cannot reorder case features or reinterpret a recovery field used by the workbench. The manifest supplies the link from a displayed case to the exact source record, while the contract digest ties that record to the admissible operating domain.

The desktop and browser paths have different responsibilities. Python is used for full scientific recomputation, model fitting and the committed bake; TypeScript serves the interactive circuit with constrained local state changes. A browser response may be deterministic and numerically close to Python, but that does not make it an observation. It remains a local calculation under the same authored assumptions. The product's API serves versioned artifacts and health metadata; it does not proxy a control system or receive sensor streams. Consequently, network availability is not a model-validity test, and a successful API call is not evidence that the nominal assumptions hold at a real site.

The release chain preserves those boundaries through version checks, template alignment, static artifact verification, notebook execution and the release-specific simulation bake. A candidate is not promoted because a page renders: its engine and artifacts must first agree, and the public bundle must then be checked against the candidate identity. This report records the 0.10.010 release candidate and its completed source-derived bake. Its version record is reserved under the DOI shown on the title page. The software release remains subject to the product repository gates, publication record synchronization and deployment verification described below.

6 Methods

The following subsections describe the implemented methods and the evidence they produce. They distinguish equations used to compute a circuit response from statistical procedures used to evaluate a learned approximation. Every comparison is conditional on the authored cases, contract, seeds and declared acceptance criteria. Model names denote implementations in this repository; the equations below do not imply that their parameters were calibrated on a plant.

6.1 Bond work-index estimate and applicability

The equations are used as engineering estimates and interpreted against the industrial work-index procedure [3, 8, 33, 32].

$$E_B = W_i \left(\frac{10}{\sqrt{P_{80}}} - \frac{10}{\sqrt{F_{80}}} \right). \tag{1}$$

$$W_{i,\mathrm{op}} = \frac{E_{\mathrm{op}}}{10/\sqrt{P_{80}} - 10/\sqrt{F_{80}}}, \qquad \eta_B = \frac{E_B}{E_{\mathrm{op}}}. \tag{2}$$

For feed and product with distributions rather than single measured percentiles, the percentile is obtained by mass interpolation on the documented grid. If $F(d)$ is cumulative passing, the selected size solves $F(d_p) = p$. The mill balance then uses the full distribution while the Bond diagnostic uses only F_{80} and P_{80}; this difference in information explains why their outputs need not move in lockstep. The reported efficiency ratio is defined as $\eta_B = E_B/E_{op}$, with the report retaining the numerator and denominator so that a ratio above or below one cannot hide a changed circuit boundary. A missing percentile is a refusal, not an arbitrary extrapolation.

The approximation is most useful for comparing a declared size reduction under a consistent test convention. It becomes unreliable as a stand-in for circuit simulation when feed composition, recycle, liberation or power limit changes. The method is included because engineers recognize it and can check the arithmetic; OreFlow's stronger process question requires the distributional model described next.

Bond's work-index relation is applied to the solved feed and product size distributions, rather than to a single visual setting on the workbench. The characteristic sizes F_{80} and P_{80} are the sizes below which eighty percent of the corresponding solids mass passes. With sizes in micrometres and work index in the contract's conventional units, the specific energy estimate is

$$E_B = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right).$$

The equation is an empirical comminution relation. W_i is not a universal ore constant inferred by OreFlow: each case carries an authored value or a documented source, and the result inherits that provenance. The model's purpose here is to expose the familiar inverse-root dependence and compare it with the energy accounting of the closed-circuit engine. It cannot resolve the full size-by-mineral breakage distribution, recirculation or installed-power clipping by itself. It is therefore reported next to the population-balance mill result, never added to it as another energy contribution.

The operating work index reverses the same relation using the observed engine energy and the solved size reduction. This makes it possible to ask whether the simulation's net circuit energy is consistent with its selected size range under the Bond convention. The efficiency ratio compares the required estimate with that operating value. A ratio is interpretable only when the reference size convention, unit conversion and circuit boundary match; it is not a standalone equipment-efficiency measurement. Bond's standard-circuit assumptions exclude several effects represented elsewhere in the engine, including classification recycle and power-limited coarse product, so discrepancy is expected when these boundaries differ.

Rittinger's surface-area law and Kick's size-similarity law are included as alternative functional relations. OreFlow calibrates both through the declared reference reduction so that all three return the Bond result at that point. Away from the reference, their curves diverge because they encode different size dependence. This calibration is pedagogical and does not establish that the three laws are interchangeable or that one is universally more accurate. The meaningful comparison uses the same feed, work index and size interval, records the reference point, and reports divergence as a function of product size. If a case enters an installed-power branch, an energy-only law cannot claim that the nominal target was achieved; the solved P_{80} must be used.

The calculation uses fresh-feed tonnes as the denominator for circuit specific energy. The mill's energy per pass is applied to its actual circulating mill feed, which can exceed fresh feed. The report's operating value is therefore reconstructed from mill power and fresh-feed throughput, not copied from the per-pass quantity.

Crusher work is accounted separately with the crushing work index. These boundaries avoid double counting and explain why a familiar Bond estimate need not equal the complete circuit KPI. The oracle tests compare a published worked example with explicit tolerances and preserve any miss; agreement at one example is a calibration check on implementation arithmetic, not universal physical validation.

For a new dataset, the method requires mass-weighted size distributions at feed and product, a work-index test with its test protocol, throughput, and an explicit circuit energy boundary. Sieves reported on a volume basis must not be treated as mass fractions without a conversion. Missing coarse or fine tails can move P_{80} materially, so the measured size range and extrapolation rule must be stated. A developer can call the energy-law function with the contract-normalized sizes, then reconcile its estimate against equipment power and the independently closed engine record. Uncertainty should vary the measured size distribution and test-derived work index together where their errors are correlated. A single nominal W_i and a precise calculator output do not quantify that uncertainty.

6.2 Size-by-mineral population balance and closed circuits

The size-class balance and liberation framing follow continuous grinding, quantitative liberation and classifier treatments [11, 14, 22].

$$\frac{d\mathbf{m}}{dt} = (\mathbf{B} - \mathbf{D})\mathbf{m} + \mathbf{f}. \tag{3}$$

$$\mathbf{s}_{\text{return}} = \mathbf{C}\mathbf{s}_{\text{mill}}, \qquad \mathbf{r} = \mathbf{f}_{\text{fresh}} + \mathbf{s}_{\text{return}} - \mathbf{s}_{\text{product}} - \mathbf{s}_{\text{tail}}. \tag{4}$$

Algorithm 1 Solve and audit a closed circuit

Require: Case contract, bounded operating point, solver tolerances
1: Validate units, bounds, mineral fractions, and circuit topology
2: Initialize recycle streams from the declared operating state
3: **repeat**
4: Apply breakage and unit transformations to size-by-mineral streams
5: Route classified products and update recycle streams
6: Check nonnegativity, water balance, and component mass residuals
7: **until** state change and residual are within declared tolerances
8: Return the trace only if all closure and finiteness checks pass

The operator solve also supplies a numerical stability diagnostic. The linear system $Ap = f$ is accepted only when its solution is finite, nonnegative within numerical tolerance and satisfies the reconstructed residual $\|Ap - f\|_\infty$ below the engine tolerance. The condition number is informative for a new size grid: near-singularity magnifies small changes in classification return into large changes in predicted size fractions. The current artifact does not claim a global condition-number bound, so the refusal and residual checks are essential safeguards.

The solved cut must also preserve the declared design load. The scalar root function compares the generated circulating load with its target and is bracketed over the contract interval. When no sign-changing bracket exists, a root-finder cannot justify an extrapolated cut; the engine should report that domain failure. Root convergence for these scalar design equations is distinct from convergence of the mill matrix solve. The pipeline records both outcomes so a coarse returned product is not mistaken for a silently converged target.

The engine represents each stream by a vector over 63 size classes for every declared mineral. Crushing and grinding are linear transformations of these vectors for a fixed operating point, while recycles introduce a closed-loop solve. For one mineral, let f be the fresh feed vector, B a lower-triangular breakage operator whose columns distribute a broken particle to finer sizes, and C a diagonal classification return operator. The crusher product is obtained by solving the classification and breakage relation; the mill's closed-circuit balance has the general form

$$[T^{-1}(e) - C]\, p = f,$$

where $T^{-1}(e)$ is the inverse mill transfer at energy per pass e. The matrix structure preserves mineral identity while allowing each mineral's density, breakage and liberation attributes to alter its path through the circuit. The operator is evaluated for each mineral and uses the same size grid, so summing across minerals yields the solids stream used for grade and throughput.

The breakage distribution and selection function determine different physical effects. The breakage matrix says where mass goes after a breakage event; the selection function describes the energy-specific rate at which each size class breaks. Their product constructs the breakage operator. The three perfectly mixed mill compartments use declared volume fractions, and their combined inverse transfer is expanded as a cubic in energy and the breakage operator. This approximation is a specified model, not a numerical solution of a fully resolved residence-time distribution. It is useful because it captures a size-dependent response and remains computationally tractable for repeated circuit solves.

Classification returns coarse material according to a partition curve. The water bypass and corrected cut create a size-dependent return fraction; mineral density modifies the cut so that dense particles can report coarser than lighter particles at the same nominal size. This feedback changes mill feed, circulating load and power. OreFlow solves the cut and energy conditions jointly with scalar root methods. When an installed-power constraint becomes active, it solves for an attainable coarser product rather than reporting an impossible target. The result records the power-limited branch so that a viewer cannot confuse a requested target with the solved product.

For a closed circuit, the balance is a fixed point: the underflow generated by classification returns to the mill, and the new mill product changes the next underflow. The solver must converge both the size distribution and scalar design conditions. A singular or ill-conditioned transfer matrix, a circulating load above its declared bound, or failure of the scalar bracket produces a structured refusal. These are meaningful model-domain outcomes and are not repaired by clipping a stream to make the arithmetic continue. Numerical tolerances and maximum iterations are taken from the engine implementation and are exercised by tests against balance residuals and known records.

A useful audit starts with a small hand-computable vector and verifies that each operator preserves nonnegative mass and the total mass expected under its declared yield. One can then compare the open-circuit transform against the closed-loop solve as return fraction tends to zero. In the full model, mineral-specific classification means the bulk size distribution cannot be substituted before the loop is solved; doing so would erase density separation. The engine therefore retains vectors until streams have passed the unit boundaries. Only then are total solids and elemental grades derived.

The population-balance calculation does not model transient load, liner wear, breakage changes with time, particle-particle interactions, or a calibrated mill survey. Each case's breakage and selection parameters are authored or sourced with their evidence status. A sensitivity sweep can show how a parameter changes the solved product, but it cannot turn an assumed parameter into a measured one. Validation combines matrix/solver checks, unit balances, published reference examples and Python/TypeScript parity. These checks establish computational consistency within the implemented assumptions, not fidelity to a particular plant.

6.3 Random-forest surrogate and case-aware evaluation

The reported estimator and comparison families are grounded in tree ensembles, boosting, Gaussian-process prediction and neural fitting [5, 7, 24, 17, 13].

$$\hat{f}_{\mathrm{RF}}(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} h_t(\mathbf{x}). \tag{5}$$

$$\mathrm{RMSE}_{\mathcal{F}} = \sqrt{\frac{1}{|\mathcal{F}|}\sum_{i\in\mathcal{F}}(y_i - \hat{f}(\mathbf{x}_i))^2}. \tag{6}$$

The primary held-out score is computed as $\mathrm{RMSE} = \sqrt{n^{-1}\sum_i (y_i - \hat{y}_i)^2}$, where the rows i belong only to the

Algorithm 2 Fit and score a case-aware surrogate

Require: Authored design rows, target, model configuration, held-out cases

for each held-out case **do**
    Exclude all its rows before fitting any preprocessing
    Fit preprocessing and model on remaining case groups
    Predict every held-out row and apply the declared observable-range diagnostic
    if the fold passes the diagnostic **then**
        Store RMSE, MAE, and test-fold R^2 with the fold denominator
    else
        Store a rejected status and null score metrics
    end if
end for
Summarize only the explicitly named admissible cohort

declared test fold. The held-out coefficient is $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$, with the test-fold mean in SST. It can be negative; in such a fold the predictions are worse than the constant test-mean benchmark. Neither score is averaged across target units. The recovery target is in percentage points, log upgrade is dimensionless, and energy is in kWh/t.

A forest's tree spread is not used as a calibrated uncertainty interval. Prediction error is assessed against exact engine outputs on the reserved rows, while the frozen GP lane reports its own interval coverage. These distinct estimands prevent an ensemble heuristic from being mislabeled as a probability statement. The published reference implementations run on the same feature matrix and fold boundaries; the comparison is not copied from literature tables.

The learning lane predicts three outputs from a fixed vector of process and ore features. The design contains 256 engine-generated operating points per case. A row is an exact engine label for the authored circuit state; the feature map includes normalized throughput, target size, work index, water and mineralogical descriptors, plus circuit-presence indicators. Case identity and group identity are not supplied as categorical shortcuts. This design asks whether the measurable engineered features carry transferable information, rather than whether a model can memorize the case name.

The random forest prediction is the mean of the trees' leaf outputs. With trees $b = 1, \ldots, B$, the fitted response is

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x).$$

The ensemble captures nonlinear interactions without requiring a linear response surface, but each leaf prediction is built from observed training targets. The finite training range therefore limits extrapolation, and tree disagreement is not automatically a calibrated predictive interval. OreFlow compares the forest against ridge, histogram gradient boosting, Gaussian process and a multilayer perceptron under identical split definitions. These alternatives represent different inductive biases, not a search over every possible model family.

The interpolation split retains rows from each case in training and test. The case holdout removes all rows for one case. The ore-group protocol removes related cases together; otherwise a sibling ore could leak distinctive mineral properties into training and make a nominal "new ore" test answer a narrower interpolation question. Every fold performs scaling inside the training partition. RMSE is computed in the physical target units and R^2 is compared with the held-out mean baseline. Negative R^2 is retained. If a fold fails the predeclared gross-range admissibility diagnostic, its error is null, not a large number chosen after seeing the residual.

The method's sensitivity is principally to the unit held out, the sample design and the response scale. A random split can make an interpolation result look strong while leaving the case or group challenge weak. Group folds have different row counts and target variance, so an average fold R^2 is not equivalent to a pooled prediction score. For neural fits, the saved seed outcomes expose training instability; the forest is

deterministic under its declared random state but still depends on its sample and hyperparameters. Feature scaling, tree depth and minimum leaf size can change fit complexity. This report uses the settings recorded with the learning artifact and does not tune against a held-out fold.

The separate frozen-model challenge evaluates already exported artifacts on a fresh seed within the same authored case families. That is not a new-case transfer test. Neural outputs are inverse-scaled with the exact released target order. An autoencoder reconstruction score is compared with its frozen threshold, and a separate GP supplies mean and standard deviation for interval coverage. The neural prediction is not substituted as the GP interval centre. Controlled direct-field shifts beyond global case bounds are synthetic probes; they test whether the guard detects those vectors, not whether a real plant can occupy them. Accepted shifts remain false negatives in the denominator.

For reuse with measured data, the response and feature definitions must first be mapped to an explicit schema, with ore, production-period and time identities retained for splitting. Replicate rows and consecutive periods cannot cross a holdout boundary when the target is future operation. Preprocessing is fit only on training data. Report error by target and by held-out group, include the number of attempted, scored and refused folds, and compare with a simple baseline. The learned lane is an approximation to the authored engine; it is not a substitute for exact process simulation or an empirical plant model unless new data, calibration and prospective tests support that claim.

6.4 Bond alternatives and energy-law comparison

The three energy-size assumptions are compared at shared inputs and a declared common anchor [3, 33, 32, 8].

$$E_R = K_R \left(\frac{1}{P} - \frac{1}{F} \right), \qquad E_K = K_K \ln \left(\frac{F}{P} \right). \tag{7}$$

$$E_B = W_i \left(\frac{10}{\sqrt{P}} - \frac{10}{\sqrt{F}} \right). \tag{8}$$

Rittinger's surface-area hypothesis can be written $E_R = K_R(1/P - 1/F)$, while Kick's size-similarity relation is $E_K = K_K \log(F/P)$. For the shared reference reduction (F_r, P_r), the constants are set by $K_R = E_B(F_r, P_r)/(1/P_r - 1/F_r)$ and $K_K = E_B(F_r, P_r)/\log(F_r/P_r)$. The fitted crossing is therefore a chosen anchor, not an independent prediction check. These formulae separate the assumptions: linearity in inverse size, logarithmic similarity, and Bond's inverse square-root difference.

The equations are evaluated only for $0 < P < F$ and a positive work index. A degenerate reduction would make the calibration denominator zero, and the code rejects it. Away from the anchor, the ratio E_R/E_B and E_K/E_B is the useful dimensionless comparison. This ratio says how alternative assumptions diverge over the sampled size range; it does not say which law is empirically best.

For a new ore, estimate W_i with a documented test and hold out size reductions for validation. Fitting the scale and evaluating at the same anchor is not validation. A robust comparison should include repeated test work, uncertainty in the percentile estimates, and a held-out feed class. OreFlow's current agreement at one worked example only verifies implementation at that example.

The energy comparison fixes feed size, product size and work index across Bond, Rittinger and Kick. Bond's inverse-square-root formulation supplies the reference value at the calibrated size reduction. Rittinger's relation makes energy proportional to newly created surface; Kick's relation uses a logarithmic size ratio. In OreFlow, two scale constants are selected at a declared reference pair (F_r, P_r) so that the alternative predictions intersect Bond there. Away from that pair, the slope rather than the reference level separates the curves. This design makes a comparison of functional assumptions visible without pretending that the constants were independently fitted to plant tests.

The curves have an important dimensional caveat. Bond's empirical constant carries the units needed by the chosen size convention; the alternative constants also inherit their calibration scale. If one changes micrometres to millimetres without transforming constants, apparent divergence is a unit error. The code converts at the method boundary and tests the reference equality. The report's plotted x-axis is a solved

product size, with the same feed size for every law. The response is a law estimate, not mill power measured from equipment. When a closed-circuit engine output is shown alongside, the reader must account for recycle and actual mill feed before comparing it with a fresh-feed specific-energy value.

The method does not adjudicate a universal winner. The Bond relation embeds a conventional test and reference circuit, while Rittinger and Kick are idealized size laws. Their divergence is informative about model-form sensitivity. Application to coarse crushing, ultrafine grinding, mixed mineral breakage or a power-limited mill needs evidence that is absent from the calibration point. The right extension is a set of size-resolved tests with replicate energy measurements and declared apparatus boundaries. Fit the constants on a training set and hold out an entire ore or size regime; otherwise matching the anchor point creates a tautological "validation."

The code reports Bond required energy, operating work index and efficiency ratio together. The operating index is reconstructed by reversing the relation on the engine's actual solved size reduction; it can differ from the authored case work index because circuit energy includes crusher, recycle, installed-power clipping and denominator conventions. The efficiency ratio is therefore a within-model diagnostic. It is not an industrial plant efficiency certification. A test report should retain throughput, measured power, size distributions, moisture basis, and whether the denominator is fresh feed or internal mill feed.

For a developer extending this comparison, add a candidate law behind the same method interface, assert dimensional conversion and exact agreement at its declared calibration point, then render its curve for all cases without changing the existing formulas. The result artifact should include formula identity, parameter provenance and solved sizes. A missing or invalid size distribution should return a refusal. Do not catch an exception and substitute zero energy, since zero would pass into an optimizer as a highly attractive but physically meaningless candidate.

6.5 Five flotation kinetic forms and bank projection

The fitted curves and bank projection are interpreted with published flotation kinetics, bubble-area, entrainment and particle-size evidence [23, 29, 9, 10, 26, 28].

$$R_1(t) = R_\infty(1 - e^{-kt}), \qquad R_{\mathrm{fs}}(t) = R_\infty[1 - fe^{-k_f t} - (1-f)e^{-k_s t}]. \tag{9}$$

$$R_{\mathrm{str}}(t) = R_\infty[1 - e^{-(kt)^\beta}], \qquad R_{\mathrm{bank}} = 1 - \prod_{j=1}^{N}[1 - R_j(\tau_j)]. \tag{10}$$

For a first-order batch model, the unfloated fraction is $U(t) = e^{-kt}$ and recovery is $R(t) = R_\infty(1 - U(t))$. The fast/slow form mixes two survival terms, $U(t) = fe^{-k_f t} + (1-f)e^{-k_s t}$, with $0 \le f \le 1$ and nonnegative rates. A gamma distribution of rates yields a heavy-tailed survival curve, while a compressed exponential $R(t) = R_\infty[1 - e^{-(kt)^\beta}]$ uses shape β to alter early and late response. Klimpel's form adds a rate-limiting response; each fitted family has a different parameter interpretation and cannot be compared solely by parameter magnitude.

The fit minimizes a declared residual objective on the common virtual batch time grid. If R_i is the reference recovery and $\hat{R}_i$ the model value, the reported RMSE is $\sqrt{n^{-1}\sum_i (R_i - \hat{R}_i)^2}$. Bank projection then propagates survival through each mixer with its own residence time. The batch residual and the projected bank error are separate checks: a fit may reproduce observed batch points yet extrapolate the late-time tail poorly. The exact distributed-rate generator provides a known synthetic truth but not external validation.

The kinetic comparison fits five forms to the same virtual batch record: first order, Kelsall fast/slow, Klimpel, gamma-distributed rates and a compressed exponential. The exact synthetic reference is generated from a declared rate distribution and noise process. This enables a controlled recovery of known parameters and a bank projection, but it does not provide an independent plant test. Fits use the same time points and objective so differences arise from model form and fitting behavior rather than a changed dataset.

For a single first-order population, remaining floatable material follows an exponential survival relation. A distribution of rates averages that survival over a nonnegative rate density. A two-component Kelsall form

uses a fast fraction and slow fraction; gamma and compressed-exponential alternatives summarize continuous heterogeneity with fewer reported parameters. Parameter bounds enforce physically admissible rates and fractions. The fit uses numerical optimization and reports convergence status and residual measures; a parameter at a bound or a flat objective may indicate that the data do not identify the chosen form.

A batch curve is not itself the recovery of a continuous bank. OreFlow projects the fitted survival through tanks in series using the declared residence time per tank and flow. If a batch rate is k and each ideal mixer has residence τ, the single-stage recovery contribution follows the competing-rate balance, and the un-recovered fraction compounds across the bank. Consequently, changing the number of tanks or total residence changes the bank response even when the batch fit is unchanged. This is a mechanistic projection under perfect-mixing assumptions, not a measured continuous-cell calibration.

The comparison reports both batch fit error and bank projection against the exact distributed-rate reference. A lower batch residual need not yield a more accurate bank result: correlated parameter errors can be amplified when survival is composed across stages. The bank projection also makes the kinetic form's tail important. A small slow fraction can dominate long residence, while a fast-only curve may fit early time points but miss the late recovery. Residual plots and parameter estimates are therefore interpreted alongside the aggregate RMSE. The rates are not fitted to a species-specific plant database, and the synthetic data generator's chosen truth is known to the experiment designer.

Sensitivity is evaluated by resampling noise and changing the time window or residence basis under a pre-declared design. Such repeats quantify variability conditional on the synthetic generator, not between-plant variability. A batch test with sparse late samples weakly identifies a slow component; a continuous circuit with recycle violates the isolated-batch assumption. Feed mineralogy and liberation can also change the effective rate distribution. The method should not extrapolate beyond the tested concentration, aeration and reagent domain merely because an exponential curve remains mathematically defined.

With new data, preserve replicate test identity, sampling time, assay error, initial mass, concentrate and tail mass, and residence-time definition. Close species balances first. Fit every kinetic form on the same training replicates, reserve whole tests for evaluation, and project all fitted curves through the same bank geometry. Report convergence, parameter-bound hits, batch residuals, endpoint recovery, bank error and the mass-balance closure. Distinguish water recovery from valuable-mineral recovery and identify whether a record is batch, continuous-cell or modeled. These checks allow the method comparison to be extended without mixing different physical estimands.

6.6 Constrained pattern search and guarded proposals

The baseline mesh search, progressive barrier and surrogate safeguards are compared with their stated derivative-free optimization frameworks [27, 1, 2, 4].

$$\max_{\mathbf{x}\in[\mathbf{l},\mathbf{u}]} J(\mathbf{x}) \quad \text{subject to} \quad g_j(\mathbf{x}) \le 0, \quad h_k(\mathbf{x}) = 0. \tag{11}$$

$$\mathbf{x}_{n+1} \in \{\mathbf{x}_n + \Delta_n \mathbf{d} : \mathbf{d} \in D_n\}, \qquad \text{an exact engine result is required before an incumbent update.} \tag{12}$$

The search's reported finite result depends on initialization, mesh, coordinate order, constraint tolerance and evaluation budget. Each decision is mapped to $[0, 1]$ using its contract bounds. Poll points outside the cube are never evaluated. A candidate with engine status refused cannot update either incumbent. For a valid evaluation, the objective is compared only after checking status and finite values. This explicit status handling prevents a failure, NaN or clipped metric from becoming a spurious optimum.

The progressive barrier tracks the best feasible objective and a least-violating infeasible incumbent. Feasible dominance is objective-based; infeasible progress is ordered by violation and objective under the current barrier. The threshold is nonincreasing, so an infeasible incumbent cannot become acceptable by relaxing the criterion after the fact. The implementation's convergence status is tied to its own stop rule. Classical pattern-search convergence theorems require assumptions on continuity, polling directions and refinement that the learned proposal insertion does not automatically establish; this report claims observed finite-suite behavior only.

Algorithm 3 Guard a surrogate proposal with exact evaluation

Require: Feasible incumbent, mesh, learned guard, exact circuit engine

```
Generate bounded poll and mesh-neighbour candidates
Rank eligible candidates using the learned objective estimate
for each selected proposal do
    Apply reconstruction and uncertainty-width guards
    if both guards pass then
        Evaluate the proposal with the exact process engine
        if the result is finite, feasible under the barrier, and improves the incumbent then
            Update incumbent and record the exact result
        end if
    end if
end for
Retain ordinary exact polling and record calls, refusals, and termination
```

A useful method ablation disables the reconstruction guard and the GP-width threshold separately, while keeping proposals, objective, starts and budget fixed. It should report exact calls, rejected candidates, endpoint feasibility and same-start differences, with repetitions across declared cases. The existing 72-pair evidence tests the integrated screen against the unscreened baseline and gives a negative compute result; it does not identify which filter component caused the extra calls.

The optimization method is a generalized pattern search with a progressive feasibility barrier. Its decision vector is normalized to the unit cube of the case-specific operating contract. At each mesh, fixed coordinate directions generate trial points around the feasible and least-violating incumbents. The process engine evaluates objective and constraints. Contract-bound violations are rejected by an extreme barrier. Grade, power and water remain relaxable process constraints whose violations are aggregated as

$$h(u) = \sum_k \max(0, -s_k(u) - \epsilon)^2,$$

where s_k is a normalized constraint slack and ϵ is the declared feasibility tolerance. Feasibility and objective improvement are separate: an infeasible point can reduce violation without becoming the best feasible solution.

Each start begins with mesh $\Delta = 1/4$. A dominating trial may expand or retain the mesh according to the recorded outcome; an unsuccessful poll contracts it. The search stops at the minimum mesh or at its distinct-evaluation budget, with a reason stored in the result. A budget stop is not convergence. Up to six starts are evaluated and duplicate starts are removed. The engine reevaluates the selected terminal point from a fresh state before it is reported. If no feasible start exists, the record retains an infeasible status and the least-violating point rather than fabricating an optimum. The objective weights recovered payable metal and specific energy relative to the base state; the baked primary result uses the declared metal-only weight and companion weight-path records are separate.

The screen proposes candidates one or two mesh steps from incumbents. A frozen neural model predicts their outputs, an autoencoder guard rejects features outside its learned reconstruction threshold, and a Gaussian process rejects candidates whose recovery half-width exceeds the configured limit. Passing candidates are ranked by the neural objective, then the exact engine performs the evaluation. The GP mean is not substituted for the neural prediction, and the interval is not claimed to be calibrated plant uncertainty. The normal pattern poll remains available. This arrangement preserves exact objective values but does not by itself prove convergence of the modified finite implementation.

The experiment compares paired searches with the same case, starts, objective, bounds, tolerances and budget, differing only by screening. Distinct cached trial-state counters measure the declared workload; they exclude model fitting, inference latency, uncached audits and the later weight path. The observed 72-variant suite has 26,367 screened versus 22,487 unscreened recorded evaluations, a 17.25% increase; counts rise in 68 variants, tie in four, and fall in none. Thus the negative result directly refutes a compute-saving claim for this suite

and measure. It does not prove that every screen increases runtime or calls.

Decision agreement is reported separately from work. Both searches return feasible endpoints in 70 cases and infeasible outcomes in two; terminal outcomes match in 65. The seven differing feasible decisions have small recovered-metal differences, yet close endpoints do not compensate for added exact work. The proposal record includes 5,811 exact evaluations, of which 901 improve the incumbent, and maximum absolute recovery error of 38.158 percentage points. That maximum demonstrates that the uncertainty-width rule is not a universal error guarantee. These outcomes motivate ablations by guard and interval threshold, but any new threshold must be fixed a priori for a new paired run.

A new application should freeze case identity, starting points, decision bounds, objective weights, exact-call accounting and stopping rules before comparison. Store every proposal, evaluated or not, its prediction and exact outcome when available, guard decision, slacks, and termination status. Compare paired counts and endpoint feasibility, and instrument wall time separately. A model that rejects an invalid plant state should not be credited with savings unless those rejected calls would otherwise have reached the exact engine. The operating endpoint must be reevaluated and reviewed by the responsible process team before any plant trial; this research optimizer is not an operational controller.

6.7 Latin-hypercube propagation and Sobol decomposition

The sampling and estimator follow variance-based sensitivity analysis; the separate geomet uncertainty study motivates retaining spatial-group structure [25, 12, 13].

$$S_i = \frac{\mathrm{Var}_{X_i}(\mathrm{E}[Y \mid X_i])}{\mathrm{Var}(Y)}, \qquad S_{T_i} = 1 - \frac{\mathrm{Var}_{X_{\sim i}}(\mathrm{E}[Y \mid X_{\sim i}])}{\mathrm{Var}(Y)}. \tag{13}$$

$$\widehat{S}_{T_i} = \frac{\frac{1}{2N}\sum_{n=1}^{N}(Y_{A,n} - Y_{AB_i,n})^2}{\widehat{\mathrm{Var}}(Y)}. \tag{14}$$

The sensitivity estimator is meaningful only for independent factors and a nonconstant output over the specified box. If the total-order estimate exceeds the first-order estimate by more than its numerical tolerance, the difference is interpreted as interaction contribution, but finite-sample estimators can reverse that ordering slightly. If the response variance is effectively zero, the ratio is undefined in practice and the method reports a constant response rather than assigning a high index to floating-point noise. The bootstrap is retained as a conservative sampling diagnostic; a low-discrepancy design violates the bootstrap's independent-row premise, so the reported half-width is not a calibrated 95% interval.

A convergence study should vary both the base sample count and seed, compare index ranks and uncertainty widths, and track refused draws. Increasing N multiplies the engine budget by $D + 2$, so a higher-resolution result has a declared computational cost. The current baked $N = 256$ and stored bootstrap design define the values displayed in the application. A user-triggered seed or sample-count replay is a separate calculation and must not replace the committed result without creating a new artifact and identity.

The uncertainty lane perturbs four inputs around each case's nominal operating point: work index, head grade, liberation size and, for flotation cases, floatability. The declared half-width is uniform and independent across the sampled factors. A seeded Latin hypercube stratifies each marginal into N bins and applies a randomized permutation. This provides broad marginal coverage with fewer runs than unstructured Monte Carlo for the same sample count, while retaining reproducible finite-sample design. It does not encode correlations among ore properties, because a dependence model is not present in the contract.

For each admitted row, the exact engine recomputes a circuit state. The artifact stores the sample inputs, seed, solved/refused identity, output vector, quantiles, mean, standard deviation and constraint probabilities. Refused draws are excluded from solved-output moments but count as failures in feasibility probabilities. This denominator split is necessary: the distribution of the outputs conditional on a steady state is not the distribution of all attempted inputs. A chart showing only solved responses without the attempted count would conceal the most important domain boundary.

The sensitivity study uses Saltelli's variance-based design with base matrices A and B and one hybrid matrix per input. For response $Y = f(X)$, the first-order and total-effect estimators are

$$\widehat{S}_i = \frac{N^{-1}\sum_j f(B)_j\{f(A_B^{(i)})_j - f(A)_j\}}{\widehat{\mathrm{Var}}(Y)}, \qquad \widehat{S}_{T_i} = \frac{(2N)^{-1}\sum_j\{f(A)_j - f(A_B^{(i)})_j\}^2}{\widehat{\mathrm{Var}}(Y)}.$$

The design costs $N(D+2)$ evaluations for D inputs. OreFlow uses a scrambled Sobol sequence and bootstrap estimates from SALib. These resampled-row half-widths are conservative diagnostics for this low-discrepancy design, not calibrated confidence intervals; a separate validation on analytic test functions found substantial overcoverage. Small negative estimates and total effects slightly below first-order estimates are sampling artifacts, not negative physical contributions. When response variance is below the declared constant-output tolerance, the record says the output is constant instead of dividing numerical noise by numerical noise.

If any Saltelli row is refused, the full matrix cannot be evaluated consistently; the artifact records the refusal and omits indices. This is not equivalent to dropping that draw and recomputing the estimator. The same rule applies in the browser replay. A structural check is available: energy should not depend on floatability when that input does not enter the energy calculation, so its effect should be exactly zero under the paired design. Such negative controls reveal accidental dependencies or a variable-order error.

Interpretation is bounded by the four selected ranges, independence assumption, sample count and steady-state solver. A high total index means the output varies strongly over this designed box; it is not a universal ranking over all deposits. To reuse the method, define supported input ranges and any dependencies before generating the design, keep refused rows, use exact engine outputs, and report attempted and solved denominators. Increase sample size as a convergence study rather than relying on one seed. If domain knowledge requires correlated inputs, use a method that preserves the joint distribution and document how its estimands differ from the independent Sobol decomposition.

7 Process model

The record reproduces all 96 committed states within 1e-6 in the browser port and all 96 committed states close within 1e-9 at every unit in the Python engine. The port shares constants and is a translation check, not an independent validation of physical assumptions.

7.1 Representation

Every stream holds, for each mineral, a vector of mass flows (t/h) over one grid of 63 size classes on a fourth-root-of-two progression from 150 mm to a pan below 3.24 µm, and a water flow. Assays are computed from mineral masses and the minerals' element contents (from their formulas and the IUPAC atomic weights); no grade is stored, so a grade changes only when a mass does. Liberation is described per valuable mineral by size (King 1979): a liberated fraction, and composites whose content and host are declared.

Each stream is represented by nonnegative mass rates indexed by mineral and size class, alongside a separate water rate. The representation retains class fractions until all units in a circuit have been evaluated; converting to a single bulk size or head grade before classification would erase mineral-specific partitioning. Mineral assays are derived from elemental composition and mineral fractions after the mass streams close. Every input grid is ordered and checked for duplicate or invalid class edges. Fractions are normalized only where the contract explicitly defines a distribution, and an invalid sum is rejected with the field path. This makes the state suitable for both physical balance equations and downstream comparison of per-mineral size curves.

7.2 Comminution

The crusher is the Whiten form (Duarte et al. 2021) [6], with a strictly lower-triangular breakage matrix $\mathbf{B}$ and a classification function $\mathbf{C}$ set by the closed-side setting:

$$\mathbf{p} = (\mathbf{I} - \mathbf{C})(\mathbf{I} - \mathbf{B}\mathbf{C})^{-1}\mathbf{f}$$

The ball mill is an energy-specific population balance (Herbst and Fuerstenau 1980) in the form of the MolyCop tools [18]: three perfect mixers of volume fractions 0.70, 0.15 and 0.15 share the breakage operator $\mathbf{D} = (\mathbf{I} - \mathbf{B})\,\mathrm{diag}(S^E)$, so the mill's inverse transfer at an energy per pass e is a cubic polynomial in $\mathbf{D}$, and the closed circuit with the cyclone returning a fraction C_i of each class solves

$$\left[\mathbf{T}^{-1}(e) - \mathrm{diag}(C)\right]\mathbf{p} = \mathbf{f},$$
$$\mathbf{T}^{-1}(e) = \mathbf{I} + c_1 e\mathbf{D} + c_2 e^2\mathbf{D}^2 + c_3 e^3\mathbf{D}^3$$

per mineral. Two scalar unknowns, the energy per pass that meets the target P80 and the host cut that holds the design circulating load, are found by the Illinois method. The energy per pass is per tonne of mill feed, so the mill draws $e\,m$, with m the fresh feed plus the cyclone underflow (less what a gravity bleed takes from it); when that exceeds the installed power, the circuit runs at installed power, $e\,m(e) = P_{\mathrm{inst}}$ solved as a fixed point, with a coarser product, and says so. The reported grinding energy is that net energy per tonne of fresh feed, $\frac{e\,m}{F}$; the crushing energy is Bond's with the crushing work index, and the Bond requirement $E = 10W_i\left(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{F}}\right)$ gives the operating work index and the efficiency ratio beside it, with Rittinger and Kick calibrated to agree with Bond at a reference reduction and never added to it (Bond 1952; GMG 2021).

7.3 Classification and separation

Hydrocyclones follow Plitt (1976): a Rosin-Rammler partition

$$y(d) = R_f + (1 - R_f)\left[1 - \exp\left(-\ln 2\left(\frac{d}{d_{50c}}\right)^m\right)\right]$$

with a water bypass R_f from the water balance and a cut corrected for each mineral's density, so dense minerals classify as coarser particles and circulate. Plitt's sizing of the cyclone cluster is uncalibrated and fails its published check (section 3), so the cluster's count and pressure are reported as an estimate, never as a result. Flotation banks are perfect mixers in series; the rate constant of each particle class follows from the bubble surface area flux $S_b = \frac{6J_g}{D_{32}}$ (Gorain et al. 1997, 1999) and a size window (Trahar 1981), a composite floats at the liberated rate scaled by its exposed surface, $P_{\mathrm{comp}} = P_V\phi^{\frac{2}{3}}$ with ϕ the valuable mineral's volume share, entrainment follows Savassi et al. (1998), and cleaner and recleaner tails return to the preceding bank. Each cell's residence is its pulp volume, less the gas hold-up, over the flow that leaves it as tail, which depends on what the cell floats, so it is solved per cell as a fixed point:

$$r_j = \frac{k\tau_j + \mathrm{ENT}\,w_j}{1 + k\tau_j + \mathrm{ENT}\,w_j},$$
$$\tau_j = \frac{V(1 - \varepsilon_g)}{Q_{\mathrm{tail},j}},$$
$$R_N = 1 - \prod_j (1 - r_j)$$

Gold follows the gravity-recoverable gold (GRG) model of Laplante, Woodcock and Noaparast as Vincent (1997) states it: the gold is a GRG component with its own size distribution, from a GRG test, plus a non-GRG component; the GRG's cyclone cut takes a density correction with an exponent of 1.0, fitted to measured GRG partitions where Stokes would give 0.5; the gravity unit on a bleed of the cyclone underflow, or of the mill discharge, recovers GRG by size at a per-pass recovery $R(d) = R_{\max}\left[1 - \exp\left(-\left(\frac{d}{x_g}\right)^2\right)\right]$; and grinding moves GRG to non-GRG at a slowed breakage rate (Banisi et al.). Low-intensity magnetic drums

capture liberated magnetite and composites by their magnetite content; desliming cyclones discard slimes ahead of flotation, with the water split following the solids sent down at a declared underflow density.

7.4 Balance closure and the contract

An independent recomputation calculates, from the output streams alone, the closure of every mineral, every species and the water at every unit and for the circuit, and of every size class at the units that do not break particles. The operating point is thirteen inputs declared once, with units, per-case bounds, steps, the families they apply to, a cross-field rule and bilingual messages; the exported declaration is interpreted identically by the engine's validator, the web service and the browser, so a state is accepted or rejected with the same code everywhere. A state the engine cannot solve, such as a classifier cut at which the mill cannot draw its installed power, is refused with a reason rather than answered.

The engine records residuals before reducing the trace to headline KPIs. For each mineral, fresh input equals the sum of product, tail and declared inventory change at the modeled steady state; water is checked separately because its units and routing differ. The contract specifies relative and absolute tolerances so small streams are not judged only against a large plant-wide denominator. A refusal preserves the failing unit, residual and reason in the run record. The report evaluates closure at internal boundaries and for the complete circuit, which can detect a duplicated recycle even when the final aggregate mass appears plausible. Closure verifies the implementation's accounting identity, not the accuracy of authored process parameters.

8 Validation

Balances. All 96 committed states close within 1e-9 relative at every unit and for the circuit, as recomputed from the stored streams by the artifact checks.

The second implementation. A line-by-line TypeScript port runs the engine in the browser. Re-simulating every committed state from its own definition, it matches every metric, stream record, curve and kinetic record within 1e-6 relative, with the same flags. It is a translation by the same author that reads the same constant, atomic-weight and mineral files, so it catches translation errors and not modelling errors. An independent verification of the 0.08.001 implementation against its equations, one reviewer per part of the engine and every finding verified, found defects in both engines, among them the composite rate, the cell residence, the mill power basis and the desliming water split; each was fixed, the engine's in both languages, with a test that fails on the earlier code, and every number below is from the fixed engine.

Published examples. Each is labelled as a published example, not as plant data:

Example	Published	Engine
Moly-Cop BallSim base case (504 t/h, every published input)	net 7.71 kWh/t, gross 8.56; cut 183.3 µm; 6 cyclones at 53 kPa	net 7.30 kWh/t, 5.2% below, inside the 20% tolerance set before the first run; cut 188 µm; Plitt's uncalibrated sizing asks for 2 cyclones at 816 kPa
GMG worked examples of the operating work index	14.4 and 11.7 kWh/t	14.38 and 11.71 kWh/t
Laplante and Staunton gravity example, run like for like (80.4% GRG, the unit on 10 to 60% of the mill discharge)	GRG recovery 79.8 to 95.9%, GRG circulating load 2016 to 413%	GRG recovery 69.5 to 90.7%, GRG circulating load 696 to 84%: outside the 5-point and 35% tolerances
Zandrivierspoort magnetite (Muthaphuli 2014; a different ore; the 45 µm product is a regrind) [19]	64.9 and 69.0% Fe at 75 and 45 µm	63.8 and 67.8% Fe: a step of 4.0 points against 4.1, at levels 1.1 and 1.2 points below

The gravity example misses because the measured GRG of the case's ore (Snip) is very fine: 9 to 31% of the GRG, almost all finer than about 37 µm, leaves by the overflow; without the GRG finer than 25 µm the GRG recovery comes within 1.1 points from the 20% row on, and the circulating load stays 50 to 69% low. The magnetite example checks a direction and a proportion, not a level, because its ore is not the engine's case.

Validation is layered because no single comparison establishes physical validity. Unit tests exercise formula limits, shape constraints, finite-value checks and refusal paths. Manufactured inputs test conservation and classification behavior where an analytic answer is available. Reference cases compare named outputs with the source's published values and preserve the discrepancy rather than fitting it away. A separate TypeScript implementation is run against the same serialized inputs and its outputs are compared under declared tolerances; this is an implementation parity check, not independent plant validation. Release gates also check schema identity, units, artifact hashes, variant counts and browser/API parity. Numerical agreement at those boundaries establishes that a committed record can be regenerated and consumed consistently. It cannot establish that the steady-state model captures an unobserved plant mechanism or that the authored parameter range covers a future ore.

9 Experimental protocol

9.1 Authored circuits

Twelve authored cases in four circuit families: nine copper, nickel, zinc and refractory-gold rougher-cleaner flotation circuits (`rougher`), free-milling gold with a gravity bleed (`gravity_rougher`), fine magnetite with magnetic drums instead of flotation (`magnetic`), and phosphate with desliming ahead of flotation (`deslime_rougher`). Every parameter carries its unit and either a source or the label authored; none is fitted to a plant. Five of the twelve are copper sulphide plants on chalcopyrite, and the two gold plants carry most of their gold in pyrite (55%, beside electrum in the free-milling ore and arsenopyrite in the refractory one); the other five each have a payable and dominant carrier no other case shares.

Cases cover rougher flotation, magnetic separation, gravity recovery, leaching and desliming configurations represented by the engine. Their identifiers, provenance notes, mineralogy and equipment declarations are preserved in each case record. The nominal operating point is not treated as a measured baseline: it is an authored reference state for controlled perturbations. Variant definitions change one named operating or ore factor at a time where the contract permits, while classifier-cut modes remain separate when their input features are outside the learned screen. The case table reports the number of variants and the response quantities available for each family, so an absent KPI is not read as a zero. Group labels are retained for transfer evaluation but are withheld from the model features.

9.2 Design, features and targets

For each case a scrambled Sobol sequence draws 256 states the contract accepts over the case's whole envelope and two ore factors (liberation size and floatability within their uncertainty half-widths), except that magnetite varies liberation alone. There are 3072 states in all; the engine simulates every state. Each state is described by 22 physical features computed before simulation: the payable fraction, throughput per installed megawatt, the grind, circulating load, water, crusher setting and work index, the payable carriers' share-weighted liberation size, composite content, density, floatability and dose ratio, the rougher gas velocity, cells and volume per t/h, the gravity bleed, the desliming cut and four circuit descriptors. The case identity is never a feature. The targets are recovery (%), the log10 upgrade ratio (concentrate over head grade) and the total specific energy (kWh/t).

The design contains 3,072 simulated states, allocated across twelve cases and generated from a versioned space-filling plan. Each row retains its case, ore group and source variant outside the predictor matrix so that grouping remains possible without leaking the identity into training. Features describe feed composition, size and liberation, hardness, water and operating controls; targets are trained separately because recovery, log upgrade and specific energy have different units and error meanings. The stored design seed and feature

order are part of the artifact identity. Duplicate states are audited before splitting. These details permit another researcher to reconstruct the training table and detect changes in case balance, target scaling or group membership.

9.3 Models and protocols

Ridge (alpha 1); a random forest (300 trees; Breiman 2001); histogram gradient boosting (300 iterations; Friedman 2001); a Gaussian process with a squared-exponential kernel with one length scale per feature and a noise term, on a seeded subsample of 500 training rows (Rasmussen and Williams 2006); and a perceptron of two hidden layers of 64 units, trained by full-batch AdamW with early stopping on a 15% validation split (Loshchilov and Hutter 2019). Features are standardized on the training rows of each split. Perceptron targets are standardized on those same rows; the Gaussian process normalizes its own training responses, while ridge and the tree models fit targets in their declared units. Because the Gaussian process sees fewer rows, ridge, random forest and gradient boosting are also refitted on the same 500 rows for a separate comparison. The perceptron's ordinary scores retain the full training subset. Because one training seed is one draw, the perceptron is retrained with four more seeds for interpolation and each of the twelve case holdouts. The two multi-case ore-group fits use the base seed; no five-seed group-transfer stability claim follows from this study.

The standardization fits the training subset alone. Numerically constant training columns retain unit scale, including columns whose small positive variance comes from summation roundoff. An unseen value is therefore not amplified by division through a spurious near-zero scale. Neither held-out features nor held-out targets determine the fitted transformation. The interpolation comparison also retains a case-mean baseline fitted on each case's training rows, and an R^2 denominator based on within-case variation, separating case recognition from response prediction.

Interpolation holds out 20% of the states inside every case (2460 training, 612 test rows). *Leave one case out* holds out each case in turn; the models that predict a case never saw any of its states, but they may have seen plants on the same ore. *Leave one ore group out* holds out every plant that shares the held-out plant's payable and dominant carrier mineral: the five chalcopyrite plants together (1792 training, 1280 test rows), the two gold plants together (2560 training, 512 test rows), and each of the other five alone, where it is the same fold as leaving one case out; seven groups in all. An autoencoder (16, 6 and 16 units) trained on the features gives a guard: its threshold is the 99th percentile of the reconstruction errors of its validation rows. Its false-alarm rate is measured on the 612 held-out in-envelope states. Its headline false-accept rate uses 11,016 probes, each a held-out state pushed half a training span above the maximum of one continuous feature. Separate acceptance curves repeat those probes at six distances from 0.02 to 1.0 spans, independently above the maximum and below the minimum.

9.4 Prediction diagnostics and admissible comparisons

Shape and finiteness are checked before an error is computed. A further gross-range diagnostic compares the prediction range with the target range over the complete authored design, expanded by one observed span on each side. For reference limits a and b, the diagnostic interval is

$$[a - (b - a),\ b + (b - a)].$$

This is a post-inference diagnostic and includes held-out target values. It is not a training-support test, an operational constraint, an independent transfer validation or a rule for selecting fitted parameters. No prediction is clipped or refitted using these limits. Because the test concerns disjoint ranges, it can miss extreme individual predictions when other predictions overlap the interval; passing it does not establish physical validity at every state.

A finite transfer forecast wholly outside that interval is retained as a rejected transfer, with null RMSE, MAE and R^2, unmodified observed/predicted/reference ranges, affected-row count and model, target, protocol, fold and seed identifiers. The same check applies to every constituent case of a grouped fold, so pooled overlap cannot conceal an entirely invalid case. Nonfinite, shape, fitting and programming failures abort evaluation;

interpolation range failures also abort. Finite poor forecasts inside the diagnostic envelope retain their errors and negative R^2. Unexpected constant forecasts are identified separately from the intentionally constant baseline.

Conditional summaries report attempted, scored and rejected counts beside their error statistics. A missing admissible cohort has null metrics. Comparisons across models additionally use the intersection of folds admissible for every model, with all excluded identifiers retained. These matched summaries remain conditional descriptions of that intersection; they do not erase a model's failed transfers or rank performance over the complete planned design. Complete per-fold, per-case and per-seed evidence is retained in the release record.

10 Results

10.1 Interpolation and case holdout

Table 1. Recovery interpolation and conditional leave-one-case-out scores. RMSE is in percentage points. Case counts are scored folds out of twelve attempts; the remaining fold is retained as rejected. Case means, maxima and medians use only each model's scored folds, so their denominators differ.

Model	Interp. RMSE	Interp. R^2	Cases scored/12	Case mean RMSE	Case maximum RMSE	Case median R^2
Ridge	9.46	0.617	12	13.68	23.05	-0.127
Random forest	6.49	0.820	12	13.16	48.86	0.359
Histogram gradient boosting	4.60	0.909	12	12.66	56.61	0.512
Gaussian process	5.34	0.878	12	10.95	27.91	0.481
Perceptron	3.38	0.951	11	6.86	12.76	0.573

The perceptron has the lowest interpolation recovery RMSE (3.38 points) on the common 612-row test set. Its conditional case mean is 6.86 points over eleven scored folds, with one rejected recovery fold. Comparing that value directly with another model's twelve-fold mean would confound model error with cohort selection. Table 3 instead compares the same eleven folds for every model.

The five-seed record gives interpolation RMSE from 3.00 to 3.54 points. The first four seeds each score eleven case folds and reject magnetite; their conditional mean RMSE ranges from 6.86 to 7.63 points. The fifth scores all twelve folds, with a 217.18-point magnetite RMSE and a 25.05-point case mean. The range diagnostic admits that last forecast because its prediction range overlaps the widened reference interval. This very poor finite error remains scored. The five conditional means therefore cannot be interpreted as a stability interval over an identical cohort.

10.2 Case transfer errors

Table 2. Recovery RMSE (points) on each held-out case.

Held-out case	Ridge	Random forest	Gradient boosting	Gaussian process	Perceptron
Soft copper porphyry	12.54	11.59	9.47	11.09	5.98
Hard copper porphyry	9.62	6.72	4.82	5.08	1.72
Free-milling gold with gravity	8.55	12.48	10.53	15.83	12.76
Fine magnetite concentration	21.01	48.86	56.61	27.91	Rejected
Nickel sulphide with serpentine slimes	13.47	8.49	6.80	6.66	6.61

Held-out case	Ridge	Random forest	Gradient boosting	Gaussian process	Perceptron
Phosphate with clay slimes	15.85	10.04	10.05	10.83	8.93
Copper-molybdenum bulk flotation	9.38	6.12	4.03	6.33	1.91
Oxide copper: assumed sulphidized response	23.05	14.12	15.19	12.94	11.60
Zinc sulphide	10.87	11.88	10.18	8.49	7.26
Copper ore with clay	10.69	5.99	6.06	9.47	6.57
Low-grade copper at high throughput	9.68	6.25	4.48	6.41	2.23
Refractory gold in sulphides	19.44	15.41	13.77	10.33	9.86

The copper cases are not uniformly easy: holding out the soft porphyry costs the perceptron 6.0 points, while its error on the hard porphyry, molybdenum and low-grade copper cases is 1.7 to 2.2 points. Mixed high-clay copper is 6.6 points. For all 256 magnetite rows, the base-seed recovery forecast ranges from -195.41 to -127.94%, wholly below the diagnostic interval of -92.94 to 196.38%; the observed simulator recovery is 68.93 to 93.32%. No clipped prediction or replacement error enters Table 2. The tree models also have large finite errors on magnetite. These model errors are measured against the authored simulator, not a plant assay.

Table 3. Matched admissible recovery cohorts. Each case column uses eleven of twelve planned folds; each group column uses six of seven. Magnetite is excluded from both intersections because the neural forecast was rejected. RMSE means are unweighted fold means, in percentage points. These are conditional descriptions, not rankings over the complete planned design.

Model	Case mean RMSE	Case median R^2	Group mean RMSE	Group median R^2
Ridge	13.01	0.013	15.15	-0.638
Random forest	9.92	0.430	11.53	0.209
Histogram gradient boosting	8.67	0.581	10.87	0.365
Gaussian process	9.40	0.498	10.95	0.318
Perceptron	6.86	0.573	14.60	0.455

10.3 Whole-ore-group transfer

Seven effective ore-group folds are planned: two multi-case groups and five single-case groups. The single-case groups use their already fitted case folds. The perceptron's magnetite rejection therefore also appears in group transfer; its group summary scores six folds, while the other models score seven.

Table 4. Conditional ore-group recovery scores. RMSE is in percentage points; counts are scored groups out of seven attempts. The perceptron's one rejected group remains in the record. Table 3 supplies the common six-group comparison.

Model	Groups scored/7	Mean RMSE	Maximum RMSE	Median R^2
Ridge	7	15.99	23.05	-0.774
Random forest	7	16.86	48.86	0.129
Histogram gradient boosting	7	17.40	56.61	0.287
Gaussian process	7	13.37	27.91	0.171
Perceptron	6	14.60	46.45	0.455

The MLP scores 0.795 on the five chalcopyrite cases held out together and -10.2 on the two gold-in-pyrite cases, with recovery RMSE of 6.72 and 46.45 points respectively. Over the common six groups, gradient

boosting has mean RMSE 10.87 points and the perceptron 14.60 points. The smaller neural case-fold mean does not persist in this grouped comparison. Per-case and per-group evidence is required to locate these differences; an aggregate does not remove the rejected transfer or the poor gold-group result.

10.4 Energy and upgrade targets

The one-case-out median R^2 for total specific energy ranges from 0.652 (ridge) to 0.952 (gradient boosting); under ore-group holdout it ranges from 0.617 (ridge) to 0.907 (Gaussian process). Log-upgrade median R^2 is negative for every model under group holdout, so the recovery transfer result does not generalize to grade-upgrade prediction. The held-out protocols and target definitions remain separate in the record.

Specific energy is measured in kilowatt-hours per tonne of solids at the declared circuit boundary. The upgrade target is the logarithm of concentrate grade divided by feed grade, which reduces the scale difference between low- and high-grade cases but changes the error unit to a dimensionless log ratio. Predictions are inverse-transformed before computing grade-scale diagnostics; averaging log errors is not equivalent to averaging percentage-point errors. Recovery remains an independent target because it is a valuable-mineral fraction and does not determine grade without mass pull. The separate heads prevent an apparently good composite score from hiding a severe error in one process objective. All three target arrays remain paired with the identical held-out rows.

10.5 Interval coverage

Nominal 95% Gaussian-process recovery intervals cover 90.7% of interpolation rows and 82.9% of pooled case-held-out rows. The worst case-fold coverage is 35.5%, and four case folds cover fewer than 80% of their rows. These intervals use the GP's own mean and predictive standard deviation. Coverage is target- and split-specific; the authored-plant split does not establish calibrated intervals on real plants.

Coverage is computed from the held-out reference falling between the model's lower and upper predictive quantiles, divided by the number of scored rows in that fold. The nominal level is a design target, not a guarantee. Coverage should be read beside interval width: intervals can cover nearly every point by being too broad to inform a decision. The report uses interpolation coverage as a diagnostic and separately shows transfer-fold behavior; pooling those records would obscure changes in the test distribution. Rejected folds do not contribute to scored-row coverage and remain visible in the attempted/scored/rejected denominator. A calibration procedure would need a prospective split and recalibration data not used to choose the model or its guard.

10.6 Reconstruction guard

On the held-out interpolation experiment, the autoencoder flags 1.47% of 612 in-envelope states and accepts 15.51% of 11,016 upward-shift probes at half a training span. Its per-feature and per-distance acceptance rates remain in `learning.json`. This validation guard differs from the final guard refitted for deployment. A low reconstruction error is a screening signal, not a process constraint, a guarantee of physical safety or a substitute for the operating contract.

The reconstruction guard compares a candidate feature vector with the learned model's training representation and applies a threshold frozen from the declared calibration records. The guard is a support diagnostic rather than a calibrated probability of correctness. It can accept an unusual vector if reconstruction error is low, or reject a valid vector if the representation cannot encode a legitimate process combination. The direct-shift probes deliberately vary individual fields beyond global case ranges to measure missed detections under controlled perturbations; they are not observed plant states. The full record identifies shifted feature, magnitude, threshold, decision and exact engine response where one was requested. A new deployment would require prospective challenge cases and a documented false-acceptance cost.

11 Kinetic, optimization and sensitivity studies

The engine produces kinetic, optimization and sensitivity records for each variant. Kinetic fits, sensitivity and unscreened optimization use the mechanistic engine. Screened optimization additionally consumes the exported neural surrogate, reconstruction guard and Gaussian-process interval.

For the paired search comparison, the recorded evaluation counter counts distinct trial states in each fixed-weight multistart search's cache; a state shared across starts is counted once. Screened and unscreened runs use separate caches, and refused trial outcomes remain recorded. These counters are captured before the subsequent warm-started weight-path study. They exclude uncached base/endpoint audits, model training and screening inference, so they are a bounded search-accounting measure rather than a tally of every simulator invocation or a wall-clock benchmark.

- **Kinetic lumping.** Five lumped models fitted to a virtual batch test of the rougher feed, projected to the bank under the same residence distribution and compared with the exact bank recovery by true flotation, over 88 flotation states: mean absolute projection errors of 1.27 points for the gamma model, 1.81 for Kelsall, 2.80 for Klimpel, 4.76 for the stretched exponential and 7.56 for first order (Polat and Chander 2000; Vinnett and Waters 2025). The first-order projection falls below the exact result at every nominal state: a batch curve that is a mixture of rates is not one exponential.
- **Constrained optimization.** A generalized pattern search with a progressive barrier from six starts (Torczon 1997; Audet and Dennis 2006, 2009) finds a point within the grade, power and water constraints for 94 of the 96 variants; the two exceptions are the magnetite circuit's harder ore and higher throughput, where the grind is the only decision and the mill is already at installed power. The learned lane screens the search step (Booker et al. 1999): candidates are ranked by the perceptron's prediction and reach the engine only where the guard accepts their state and the Gaussian process's 95% half-width on recovery is within 5 points, a novelty measure for the perceptron rather than an interval on what it ranks; the poll and the optimum stay engine results. The screen did not save engine work: over the 72 screened variants it used 26,367 engine evaluations versus 22,487 without screening, 17.25% more. It saved calls in none, used more in 68 variants and tied in four. Mean absolute recovery error averaged 0.66 percentage points across the 68 variants with evaluated proposals; four variants have no proposal-error estimate rather than zero error. Across 5,811 evaluated proposals its mean was 0.71 points, median 0.48 and maximum 38.16. Both procedures had the same terminal outcome in 65 variants: 63 identical feasible decisions and two matching infeasible outcomes. Seven feasible pairs selected different decisions, with recovered-metal differences no greater than 0.002653 t/h (0.017%). Five favored the unscreened decision and two the screened decision.
- **Sensitivity.** Sobol indices of four uncertain ore properties at each nominal state (Saltelli et al. 2010; Herman and Usher 2017): floatability drives recovery in ten cases, liberation size drives concentrate grade in seven and the head grade in the other five, the work index drives grinding energy and the head grade drives recovered metal in all twelve.

12 Measured-data comparisons

The authored circuit parameters are not fitted to these datasets. Three measured-data studies are evaluated separately; GeoMet assays also supply inputs to an explicit mechanistic comparison.

Particles. The HZDR constructed-case workbook (Pereira et al. 2021; RODARE 336, CC BY 4.0) keeps its training sheet (68 008 particles with A/B labels) and its test sheet (29 147 particles with constructed probabilities and the source authors' predictions, but no observed labels). A sparse logistic model per case and a shared network (4, 32, 32, 4) are scored against the constructed probabilities:

Case	Published reference	L1 logistic	Network
1	0.0257	0.0131	0.0138
2	0.0175	0.0157	0.0181

Case	Published reference	L1 logistic	Network
3	0.0207	0.0171	0.0143
4 (28 484 complete rows)	0.0371	0.1919	0.0226

The linear model beats the network in cases 1 and 2 and fails only in case 4, where the network does not.

Locked-cycle tests. 52 GeoMet locked-cycle copper recoveries from 29 drill holes of one deposit (Hoffimann et al. 2022; archived dataset 7051975, CC BY 4.0), predicted from five assays by the training mean, ridge, a random forest and a Gaussian process, under five folds of whole holes and three spatial zones. RMSE ranges from 5.09 to 5.51 points (whole holes) and 5.15 to 5.69 (zones). A paired bootstrap over complete holes (2000 resamples) shows how little separates them. On the five fixed hole folds ridge beats the training mean by 0.42 points (95% interval 0.02 to 0.82), but that partition sits at the 98.5th percentile of 200 random hole partitions; averaged over them the gain is 0.17 points (-0.39 to 0.71) and under leave one hole out 0.22 (-0.36 to 0.79), each interval widened for the six model pairs, and under spatial zones no model separates either. The source has no grind, reagent or residence information, so it cannot calibrate the engine's controls.

The engine on the same samples. The 52 samples also run through the engine as inputs, each on its own assays in the soft copper porphyry's circuit at that case's operating point; nothing in the engine is fitted to them. The work index comes from the nearest comminution sample in the hole (42 samples) or the deposit median (10), and spans 15.2 to 26.1 kWh/t against the case's 11.0. Most of the copper is bornite and chalcocite, allocated by a sulphur-limited normative mineralogy, one choice in a family of allocations; the two minerals float at declared ratios to chalcopyrite (0.8, and chalcocite at 1.5 times bornite, a choice the GeoMet data do not bound). Every sample runs the mill at installed power, and the engine's recovery falls short of the locked-cycle test by 22.6 points on average (RMSE 24.1 points; 2.9 to 41.1 points below). That gap is mostly the host circuit's size: the samples are harder than the circuit's design ore, so at 720 t/h the product is coarse. With the circuit sized for the case's 150 µm target the gap is +0.2 points, and it runs from +10.0 to -20.8 points as the assumed laboratory grind goes from 75 to 300 µm, which the open data do not give. At each sample's own target-grind throughput it is +4.0 points, of which 3.8 come from the longer flotation residence at the lower throughput rather than from the grind. At every assumed grind the engine carries no information about which sample recovers more (Pearson r between 0.03 and 0.06). It is a comparison, not a calibration: the circuit, the breakage, the liberation and the flotation are authored for another ore, and a locked-cycle test is not a plant.

Plant hours. One iron-ore plant's reverse flotation record (Kaggle dataset 6294, CC0; 737,453 rows) [20] gives hourly laboratory silica beside 21 feed, reagent, pulp and column sensors. Dropping the 310 hours whose silica label was interpolated leaves 3,701 pairs of consecutive hours. Scored on three future windows after at least 24 hours of embargo, the previous assay alone forecasts the next hour's silica with a mean absolute error of 0.464 points, and the best sensor-only model, ridge, is 0.001 points below the training mean's 0.766. In 461 of those pairs the laboratory value is one carried over unchanged for three or more consecutive hours; without them the previous assay's error is 0.510 points, still the lowest, and ridge is 0.014 points above the training mean. The engine has no reverse cationic flotation family, so these hours are shown and never simulated.

13 Results evidence

The artifact-derived evidence is assembled below rather than manually re-entered. Figure 1 displays the case-to-case range of modeled recovery and concentrate grade; the grade axis is absent where a case has no defined concentrate-grade quantity. Table 8 preserves the corresponding case identifiers and values. Figure 9 and Table 9 show how the recovery error changes when the held-out unit changes from an operating state to a whole case or ore group. The aggregate columns use each model's recorded admissible folds, so they must be read together with the scored and rejected denominators.

Figure 2 and Table 10 expose the neural model's result for each held-out case. The red, labelled bar denotes a diagnostic rejection and is not a zero-error case. Figure 3 and Table 11 retain negative group-held-out coefficients of determination, which indicate that the model can be worse than a cohort-mean predictor.

Figure 5 and Table 12 compare batch-fit errors for five fitted kinetic forms; their bank-lumping errors are separate measures and do not imply that the lowest batch error will yield the lowest circuit error.

Figure 7 and Table 13 compare exact cached engine evaluations for paired nominal case searches. The points above the identity line have a higher screened count; they are not elapsed-time measurements. Figure 6 and Table 14 keep the external measured-sample lane distinct from the authored simulator benchmark. The offset is a model-domain mismatch to investigate, not evidence that these measurements calibrated the engine.

Figure 4 checks empirical coverage against the nominal interpolation interval level. It uses the interpolation test partition and cannot establish calibration under case or ore-group transfer. Figure 10 and Table 15 show output quantiles within authored input boxes; the intervals are not population-frequency statements. Finally, Figure 8 and Table 16 reproduce the archived particle-reference calibration, with marker size denoting test-row support. These results are limited to the source's constructed probabilities and its documented protocol.

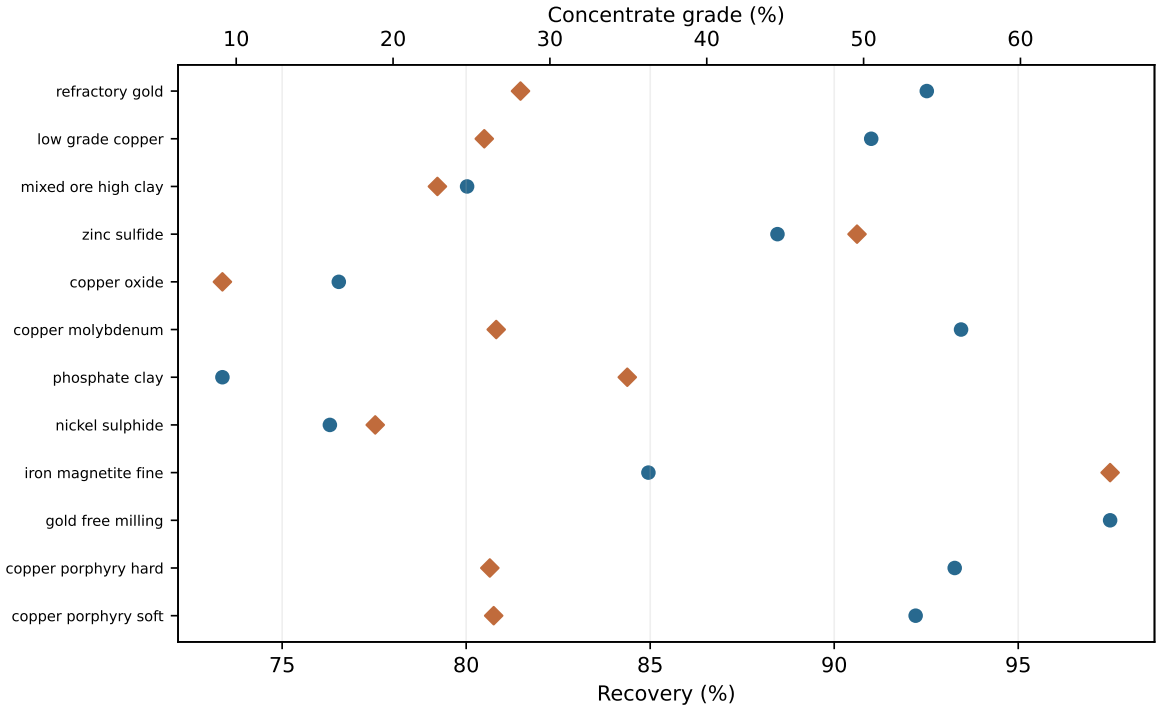


Figure 1: Modeled recovery and concentrate grade across twelve authored circuits; paired axes preserve their distinct units and the points are not plant measurements.

14 Discussion

The test bounds transfer between authored plants. A thirteenth authored plant is not a real one: the engine is a steady-state model with authored parameters, no froth stability, no dynamics and no prices, and its twelve plants share its structural assumptions. That is a feature of the design, since the teacher's answer is exact and reproducible, and a limit of the conclusion, since a real plant can depart from all twelve at once.

Four interpretation limits emerge from this simulator study. First, interpolation and transfer answer different questions: a good interpolation score coexists with a rejected transfer and a large finite error under another seed. Second, the held-out unit changes the evidence. Sharing a dominant mineral does not make whole-case and whole-group tests interchangeable; the two gold cases expose a large grouped error. Third, admissibility diagnostics change the scored denominator. A lower conditional mean cannot be presented as an improvement over a complete-design mean, and even a matched comparison remains conditional on the shared admissible

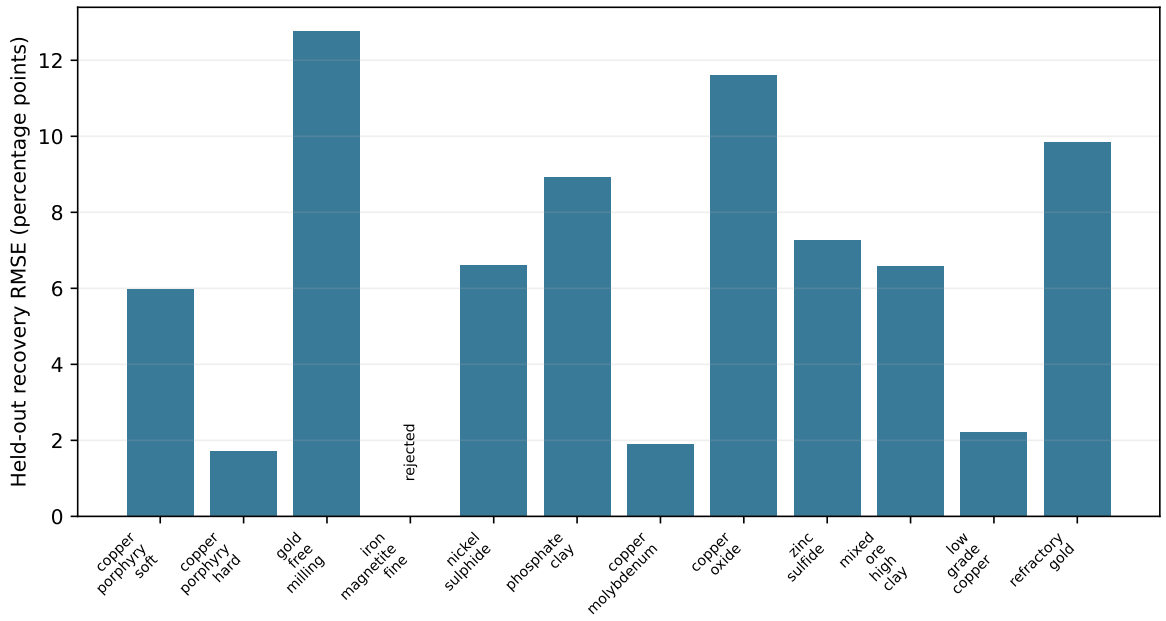


Figure 2: Neural recovery error for each held-out authored case; the labelled rejection marks a failed range diagnostic and is not assigned a numerical score.

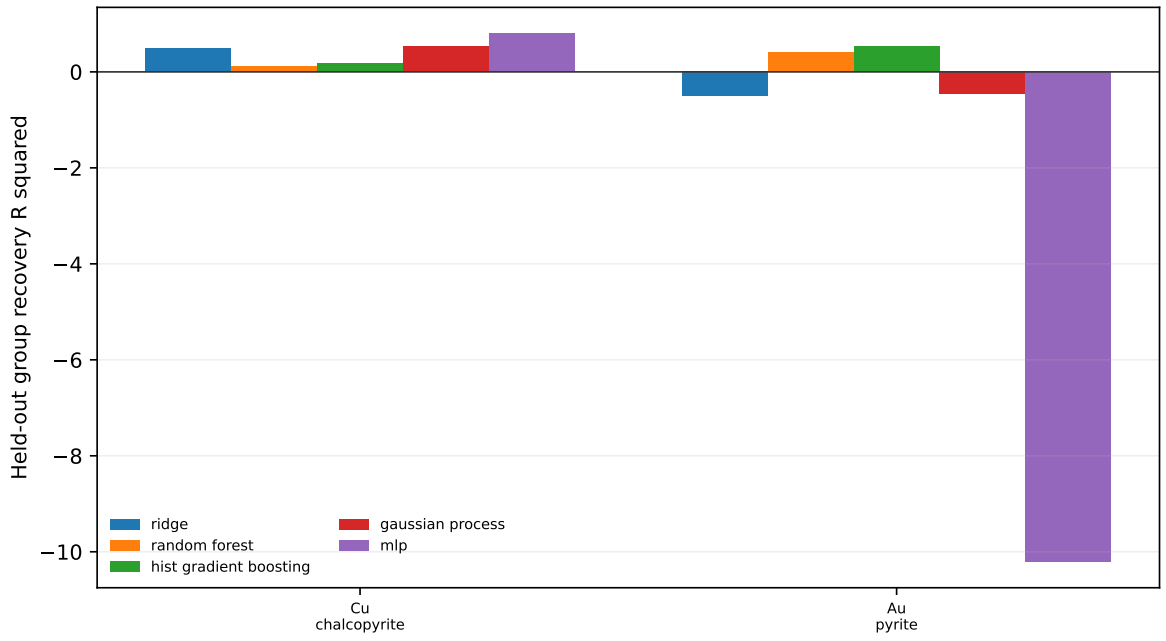


Figure 3: Recovery determination scores by held-out ore group and model, including negative values that indicate performance below the corresponding cohort-mean predictor.

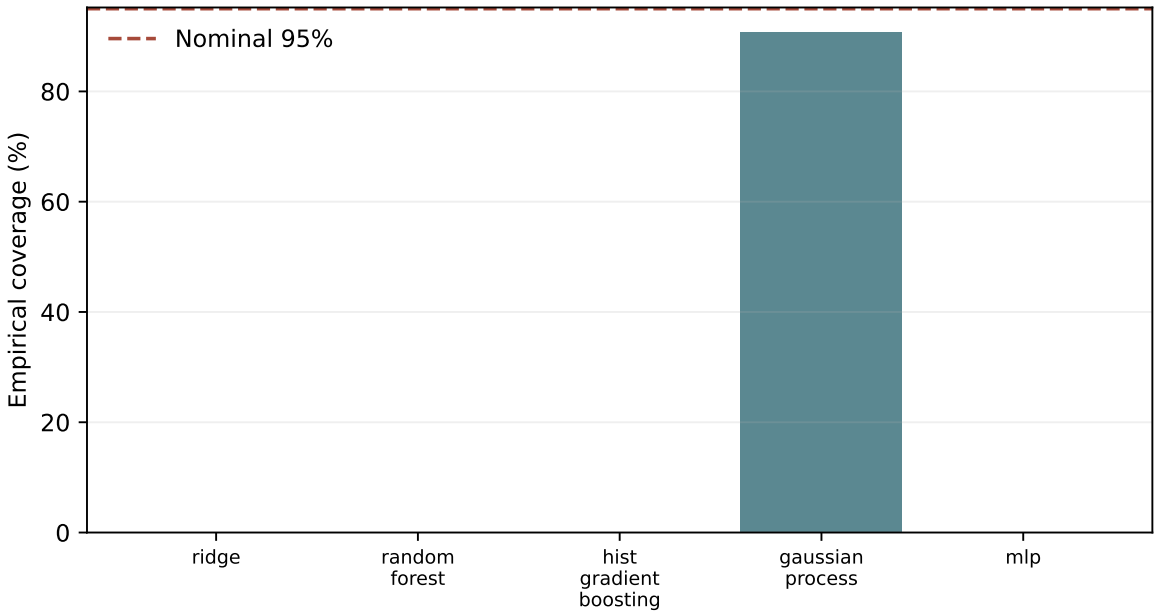


Figure 4: Empirical recovery-interval coverage on the interpolation test partition compared with the nominal ninety-five-percent reference level.

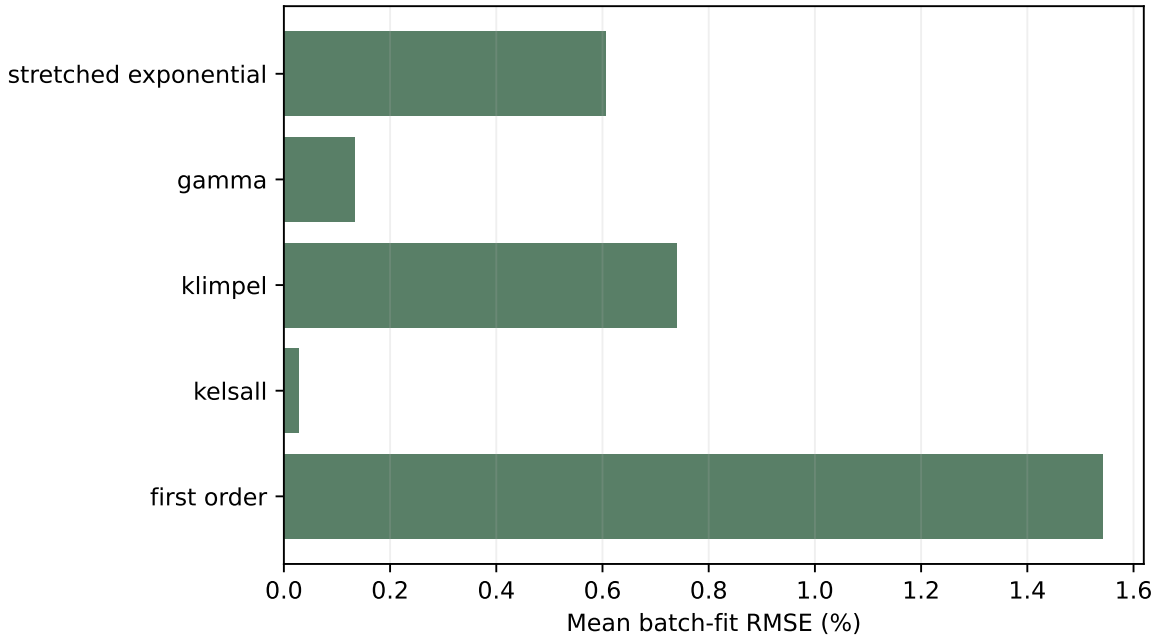


Figure 5: Mean fitted batch-recovery error for five implemented kinetic forms over the common virtual batch-record set.

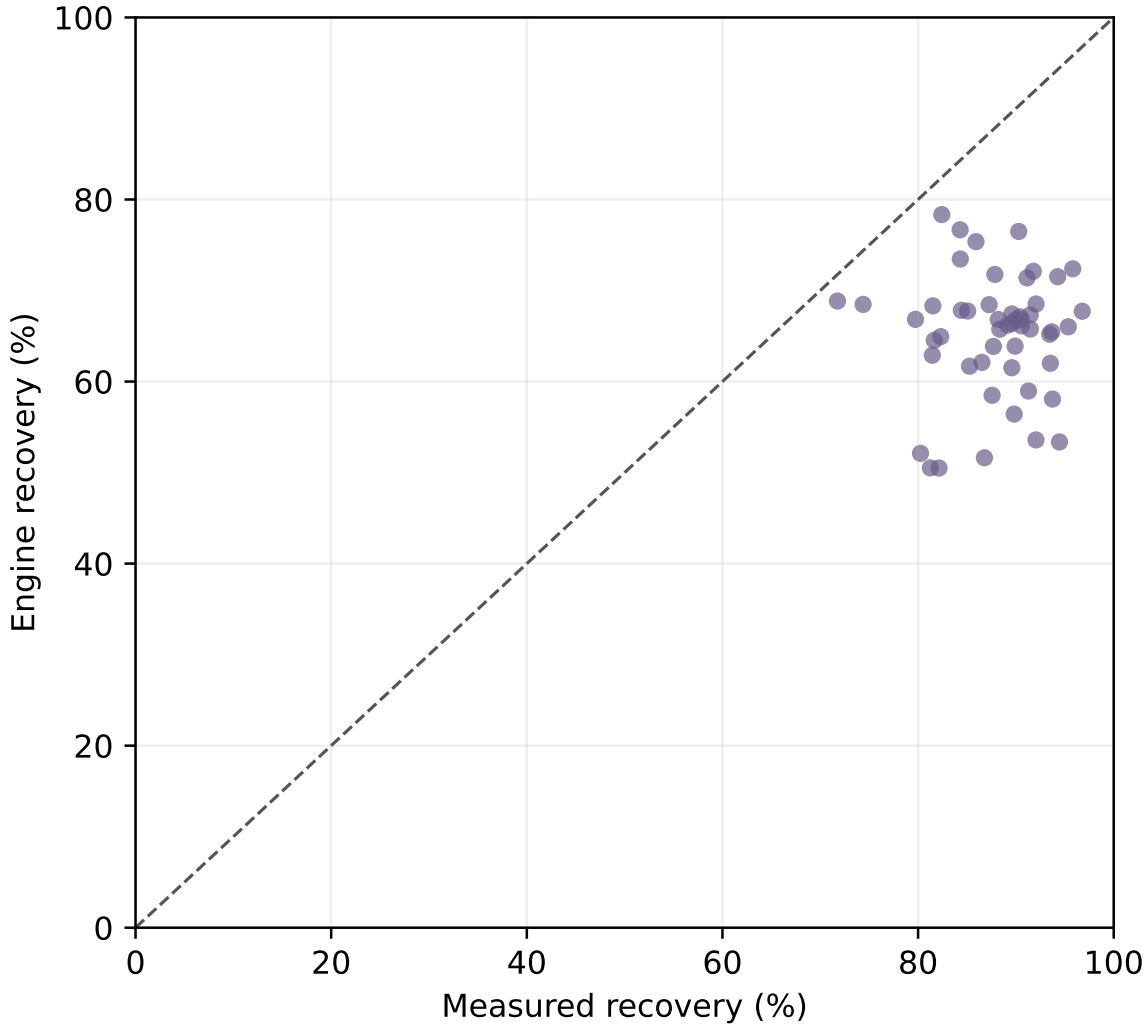


Figure 6: Mechanistic engine recovery against 52 external measured samples, with the identity line showing the direction and magnitude of the domain mismatch.

Table 8: Modeled recovery and concentrate grade for twelve authored circuits; the points are simulator outputs, not measured plant observations.

Case	Family	Recovery (%)	Grade (%)
copper_porphyry_soft	rougher	92.21	26.42
copper_porphyry_hard	rougher	93.27	26.17
gold_free_milling	gravity_rougher	97.50	–
iron_magnetite_fine	magnetic	84.95	65.72
nickel_sulphide	deslime_rougher	76.30	18.86
phosphate_clay	deslime_rougher	73.38	34.93
copper_molybdenum	rougher	93.45	26.58
copper_oxide	rougher	76.54	9.12
zinc_sulfide	rougher	88.46	49.58
mixed_ore_high_clay	deslime_rougher	80.03	22.83
low_grade_copper	rougher	91.01	25.82
refractory_gold	rougher	92.52	28.13

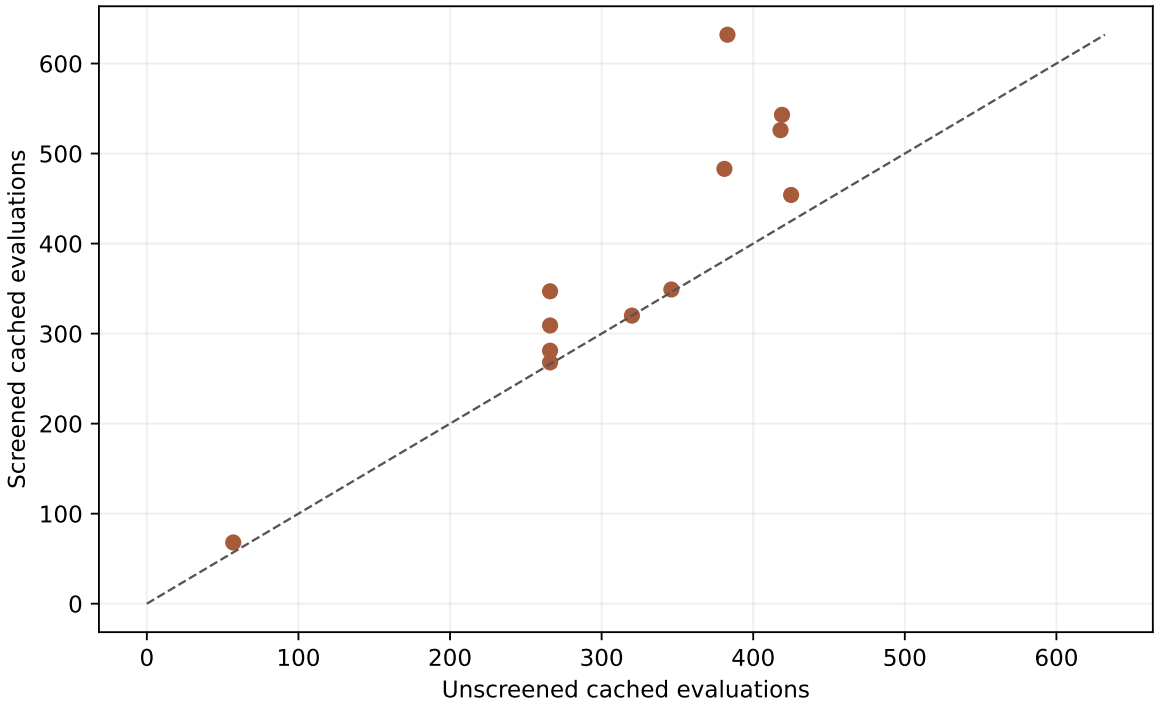


Figure 7: Paired cached exact-engine evaluation counts for nominal circuit searches, where points above identity indicate additional calls under surrogate screening.

Table 9: Recovery prediction errors by model and evaluation protocol, with attempted, scored and rejected fold counts kept beside the conditional metrics.

Model	Interpolation	Held-out case	Held-out group	Case scored/rejected	Group scored/rejected
ridge	9.46	13.68	15.99	12/0	7/0
random_forest	6.49	13.16	16.86	12/0	7/0
hist_gradient_boosting	4.60	12.66	17.40	12/0	7/0
gaussian_process	5.34	10.95	13.37	12/0	7/0
mlp	3.38	6.86	14.60	11/1	6/1

Table 10: Every held-out case result for the multilayer perceptron, preserving gross-range rejections instead of assigning them an artificial error score.

Held-out case	Rows	Status	RMSE (pp)	R squared
copper_porphyry_soft	256	scored	5.98	0.854
copper_porphyry_hard	256	scored	1.72	0.988
gold_free_milling	256	scored	12.76	-1.199
iron_magnetite_fine	256	rejected-transfer	–	–
nickel_sulphide	256	scored	6.61	0.472
phosphate_clay	256	scored	8.93	0.437
copper_molybdenum	256	scored	1.91	0.985
copper_oxide	256	scored	11.60	0.185
zinc_sulfide	256	scored	7.26	0.787
mixed_ore_high_clay	256	scored	6.57	0.520
low_grade_copper	256	scored	2.23	0.980
refractory_gold	256	scored	9.86	0.573

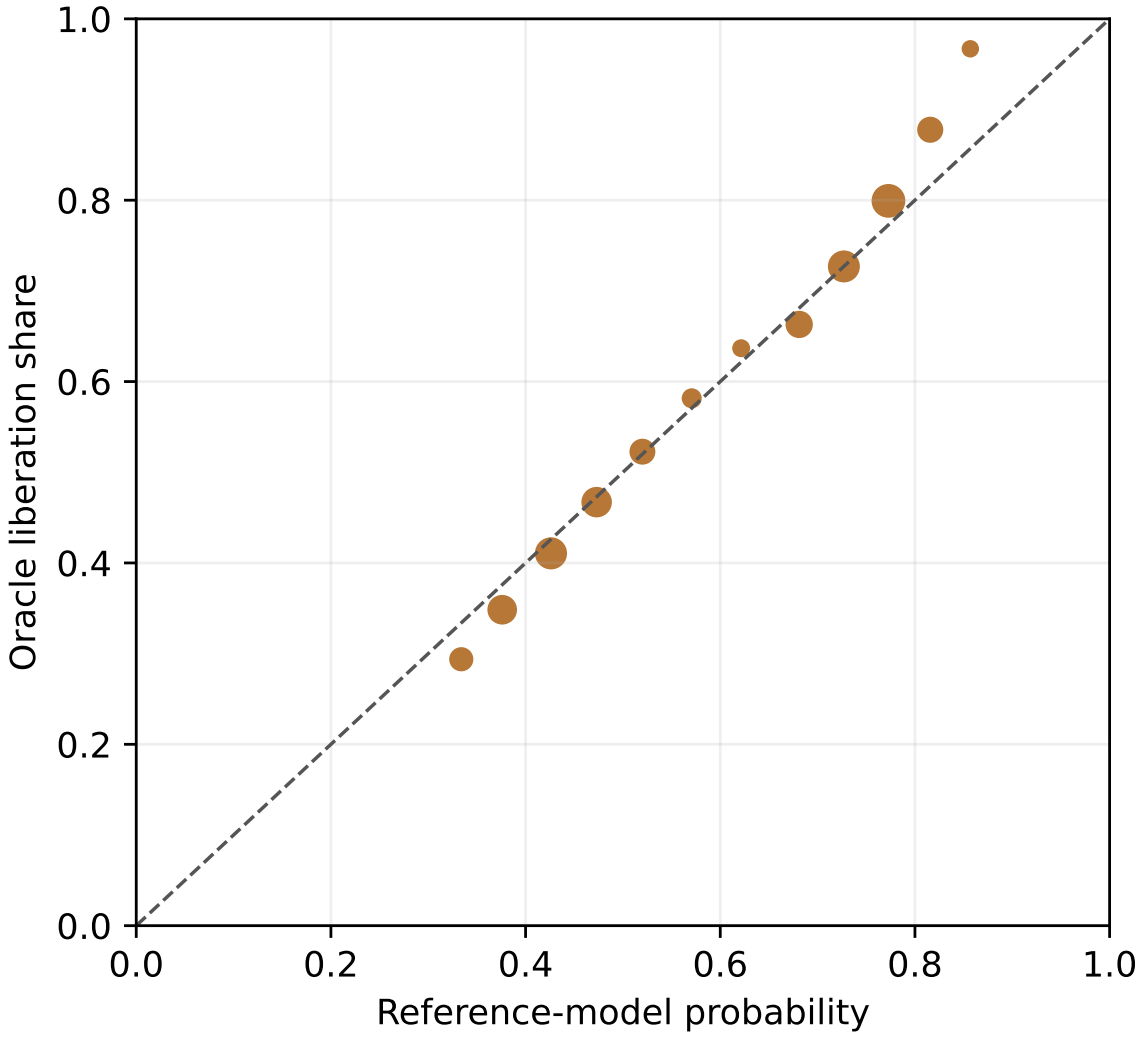


Figure 8: Archived particle-reference calibration bins against constructed oracle liberation shares, with marker area proportional to the number of test rows.

Table 11: Model scores for each held-out ore group; negative R squared values remain visible and group identity is not supplied as a predictor.

Ore group	Cases	ridge	random_forest	hist_gradient_boosting	gaussian_process	mlp
Cu:chalcopyrite	5	0.485	0.119	0.179	0.539	0.795
Au:pyrite	2	-0.502	0.404	0.528	-0.448	-10.200

Table 12: Five kinetic forms fitted on a common set of virtual batch records; the table reports fit error, circuit-lumping error and convergence.

Kinetic form	Fits	RMSE (%)	Mean lumping error (%)	Worst error (%)	Converged
first_order	88	1.5422	7.559	19.132	1.000
kelsall	88	0.0280	1.806	8.473	1.000
klimpel	88	0.7397	2.805	10.555	1.000
gamma	88	0.1332	1.266	5.070	1.000
stretched_exponential	88	0.6051	4.763	14.732	1.000

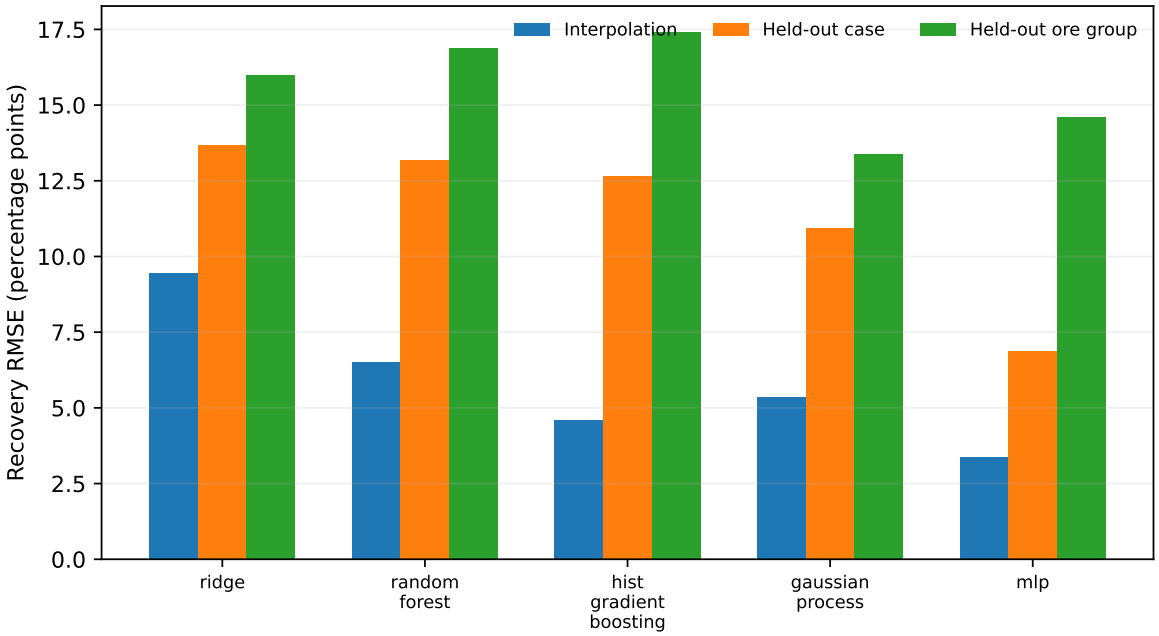


Figure 9: Recovery error for five models under interpolation, whole-case holdout and ore-group holdout, showing that each protocol estimates a different transfer question.

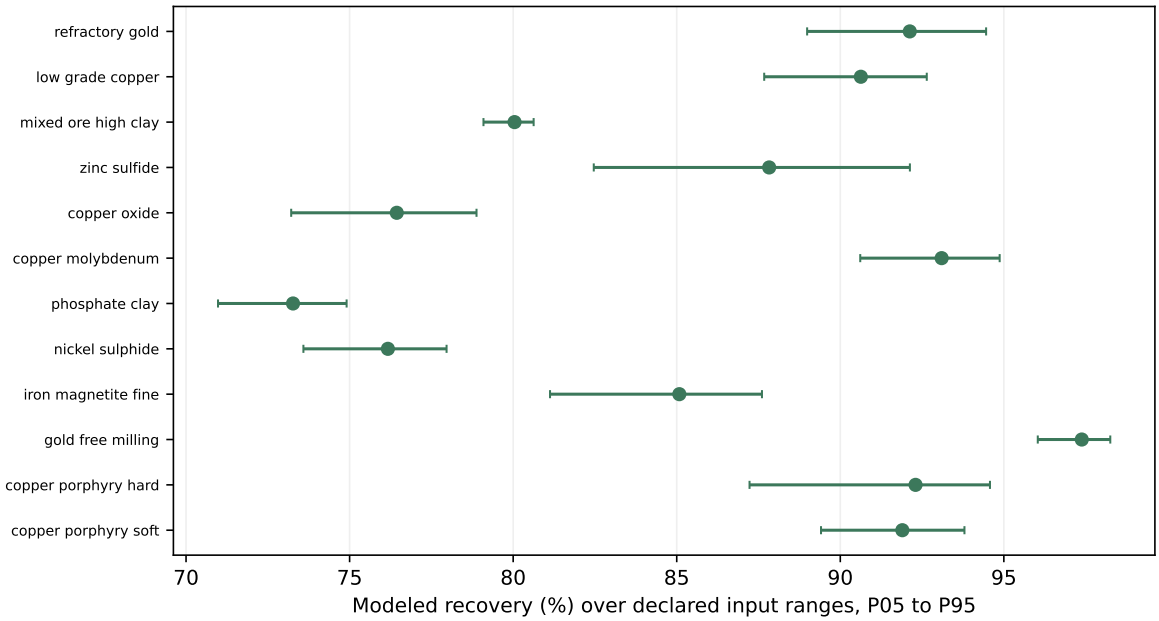


Figure 10: Median and fifth-to-ninety-fifth recovery quantiles over each authored case's declared input ranges, not a population-frequency distribution.

Table 13: Paired exact-engine evaluation counters at each case's nominal operating point; inference, audits and the later objective-weight path are excluded.

Case	Unscreened	Screened	Difference	Proposed	Improved
copper_porphyry_soft	266	309	43	72	18
copper_porphyry_hard	418	526	108	186	21
gold_free_milling	266	268	2	13	5
iron_magnetite_fine	57	68	11	28	2
nickel_sulphide	346	349	3	32	7
phosphate_clay	320	320	0	0	0
copper_molybdenum	266	347	81	108	21
copper_oxide	383	632	249	154	18
zinc_sulfide	425	454	29	81	16
mixed_ore_high_clay	419	543	124	146	17
low_grade_copper	381	483	102	173	22
refractory_gold	266	281	15	26	9

Table 14: External sample comparison of the mechanistic engine and separate learned data-lane models, with the measured sample count and error units stated.

Method	Samples	Bias (pp)	RMSE (pp)
Mechanistic engine	52	-22.65	24.14
train_mean	52	-0.03	5.51
ridge	52	-0.12	5.09
random_forest	52	-0.53	5.41
gaussian_process	52	-0.56	5.20

Table 15: Seeded recovery quantiles and feasibility probabilities for authored cases, conditional on the declared independent-factor ranges rather than plant-frequency distributions.

Case	P05	P50	P95	All constraints
copper_porphyry_soft	89.41	91.90	93.80	0.820
copper_porphyry_hard	87.23	92.30	94.58	0.539
gold_free_milling	96.04	97.38	98.25	0.820
iron_magnetite_fine	81.13	85.08	87.61	0.523
nickel_sulphide	73.59	76.17	77.96	0.789
phosphate_clay	70.98	73.27	74.91	0.836
copper_molybdenum	90.61	93.10	94.87	0.805
copper_oxide	73.21	76.44	78.88	0.875
zinc_sulfide	82.46	87.83	92.13	0.438
mixed_ore_high_clay	79.09	80.04	80.63	0.641
low_grade_copper	87.67	90.63	92.64	0.797
refractory_gold	88.99	92.12	94.46	0.789

Table 16: Calibration bins from the archived particle benchmark reference model; marker area is proportional to test rows and the benchmark protocol remains in the source record.

Case	RMSE	MAE	Bias
1	0.02572	0.01994	-0.00039
2	0.01750	0.01652	0.00043
3	0.02070	0.01766	-0.00365
4	0.03714	0.02721	0.00020

folds. Fourth, reconstruction errors miss some controlled feature shifts, so their acceptance rates need the feature, distance and tested population beside them. These observations motivate a measured study on real operating periods; they do not prove the same rankings or rates on a plant.

A calibrated study would replace the authored targets by measured metallurgy across operating periods, calibrate the engine on some periods and hold out an entire period and an ore family, and score both point error and whether the surrogate preserves the ranking of operating decisions.

The strongest product result is reproducibility of a declared computational experiment, not generalization to a mine. A closed balance is necessary for a meaningful simulator response, but mass closure alone cannot identify a wrong breakage law, liberation curve or floatability parameter. Similarly, close Python and browser outputs show that two implementations agree on the same equations and inputs; they can share the same structural error. The measured-data lane shows this distinction in practice: the engine has a sizeable recovery bias against GeoMet samples, while data-lane regressors reduce that particular error under a different target and feature contract. Those regressors do not repair the mechanistic model because they estimate from the external data lane and cannot be substituted for the circuit equations. The negative screen result is also useful: it rules out a claimed call saving under the evaluated finite settings and motivates targeted ablations before further claims.

15 Limitations

OreFlow represents a steady-state, deterministic flowsheet with authored equipment and ore parameters. It does not model plant transients, sensor delay, control logic, froth stability, stochastic equipment failure, maintenance, reagent cost, energy tariffs, concentrate penalties, water scarcity prices, capital expenditure or mine scheduling. An optimized decision in this engine therefore maximizes a declared simulator objective and satisfies declared simulator constraints; it is not an economic optimum or an operating recommendation.

The twelve cases are designed research scenarios, not a probability sample of mines. Their shared equations and parameterization make within-engine experiments reproducible, but create structural dependence among cases. The 3,072 learning states come from a designed simulator grid, so even whole-case and ore-group partitions test transfer within that authored domain. The measured GeoMet samples come from a different data lane and show a substantial mechanistic recovery bias; the missing plant controls, equipment states and process-time alignment prevent attributing that bias to one physical cause. The particle reference has constructed test probabilities rather than observed liberation labels. Neither source supplies prospective validation on a production plant.

The uncertainty calculations vary declared factors under an independent-factor design. They do not encode geological covariance, temporal drift or measurement uncertainty. Bootstrap widths quantify sensitivity-estimator variation under the chosen design, not confidence in a plant parameter. The screening experiment uses finite budgets and cached distinct-state counters. It has no wall-clock instrumentation, and the observed increase in exact calls is a negative result for this suite, not a universal claim about all surrogates or optimizers. These boundaries constrain interpretation until time-aligned plant data, prospective holdouts and an independently reviewed deployment protocol are available.

OreFlow is a steady-state deterministic simulator with authored ore and equipment parameters. It does not model plant transients, sensor delay, control logic, froth stability, equipment failure, maintenance, reagent cost, energy tariffs, concentrate penalties, capital expenditure or mine scheduling. Its optimizer therefore returns a finite simulator decision, not an economic optimum or operating recommendation. The twelve cases are research scenarios rather than a probability sample of mines, and their shared equations create structural dependence. The 3,072 learning rows test transfer within this authored domain. The GeoMet samples lack time-aligned plant controls and equipment states, preventing attribution of the large mechanistic bias to one physical cause. The particle benchmark uses constructed probabilities rather than observed liberation labels. The uncertainty analysis assumes independent factors over authored boxes and does not encode geological covariance or temporal drift. The screening counters exclude inference and elapsed time; the measured increase in exact calls is limited to this suite and budget.

16 Conclusions

OreFlow provides a reproducible steady-state process engine and an explicit set of learning, optimization and uncertainty experiments over twelve authored mineral-processing cases. Its strongest evidence is computationally bounded: the Python engine closes the modeled balances, the TypeScript workbench reproduces the committed states within declared tolerances, and the report preserves misses against published or measured references rather than treating them as calibration. None of these checks converts the authored circuits into plant-calibrated models.

The learning results show why validation unit matters. Interpolation performance can coexist with a rejected magnetite fold, seed instability and negative grouped-transfer scores for log upgrade. A score averaged over admitted folds must retain its denominator and cannot be presented as performance on the excluded ore. The optimizer comparison is also negative for its stated goal: in 72 paired target-mode variants, its learned screen increased recorded cached engine evaluations by 17.25%, with more evaluations in 68 cases, ties in four and savings in none. Similar feasible endpoints do not negate that cost result. The uncertainty analysis describes output variation inside authored independent-factor boxes; its Sobol bootstrap widths are conservative diagnostics rather than calibrated confidence intervals.

Taken together, the product is a research workbench for inspecting mechanistic assumptions and evaluation design. It is not a plant digital twin, an economic optimizer or an operating recommendation system. The next scientifically useful step is not to broaden claims from these finite records; it is to acquire time-aligned plant metallurgy, process controls and equipment data, define prospective ore and production-period holdouts, calibrate only on designated periods, and compare against untouched periods with uncertainty and a documented safety review. Until those data exist, OreFlow provides transparent computation, not a claim of industrial prediction.

17 Reproducibility

The repository is https://github.com/fsantibanezleal/CAOS_OreFlow (MIT). The engine, the bake, the records, the browser port, the service and the documentation are versioned together; the source, records and computational environment are versioned with the software release. `./scripts/setup.ps1` builds the environments and `./scripts/precompute.ps1` regenerates every record. The 0.10.010 validation record reports 4,606 s for the twelve cases on four workers, 3,350 s for the learning stage, 360 s for studies and 149 s for real-sample processing. The learning design, all fifteen evaluation jobs and all 65 neural-seed jobs were executed fresh, with no resumed checkpoints. The learned stage records CUDA execution on an RTX 4070 Laptop GPU. Its fit and evaluation concern authored simulator cases; this computation does not establish plant-calibrated performance. Independent evaluation folds and neural seed runs can execute in bounded spawned workers. External checkpoints are accepted only under the same scientific source, JSON constants, release version, case definitions, operating contract, settings, dependency versions and compute-runtime fingerprint. Each checkpoint has a verified JSON checksum and scientific diagnostics; corrupt or foreign records fail rather than being silently reused. Execution metadata distinguishes planned, computed and resumed jobs. Final deployed-model fitting and ONNX parity checks were executed in the completed learned stage. Checkpoint reuse within that identical candidate is distinct from the development-only reuse of an earlier release's learning record, which is rejected for publication. `./scripts/smoke.ps1` runs the checks: the guards, the Python and frontend suites and the build, with the counts of each release in `docs/release-verification.md`. The workbench at https://oreflow.ml.fasl-work.com runs the engine in the browser on any state of any case.

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